3.26 Determine the signal $x(n)$ with z-transform

$$X(z) = \frac{3}{1 - \frac{10}{3}z^{-1} + z^{-2}}$$

if $X(z)$ converges on the unit circle.

3.27 Prove the complex convolution relation given by (3.2.22).

3.28 Prove the conjugation properties and Parseval's relation for the z-transform given in Table 3.2.

3.29 In Example 3.4.1 we solved for $x(n)$, $n < 0$, by performing contour integrations for each value of n. In general, this procedure proves to be tedious. It can be avoided by making a transformation in the contour integral from z-plane to the $w = 1/z$ plane. Thus a circle of radius R in the z-plane is mapped into a circle of radius $1/R$ in the w-plane. As a consequence, a pole inside the unit circle in the z-plane is mapped into a pole outside the unit circle in the w-plane. By making the change of variable $w = 1/z$ in the contour integral, determine the sequence $x(n)$ for $n < 0$ in Example 3.4.1.

3.30 Let $x(n)$, $0 \le n \le N - 1$ be a finite-duration sequence, which is also real-valued and even. Show that the zeros of the polynomial $X(z)$ occur in mirror-image pairs about the unit circle. That is, if $z = re^{j\theta}$ is a zero of $X(z)$, then $z = (1/r)e^{j\theta}$ is also a zero.

3.31 Compute the convolution of the following pair of signals in the time domain and by using the one-sided z-transform.

(a) $x_1(n) = \{1, 1, \underset{\uparrow}{1}, 1, 1\}$ $\qquad x_2(n) = \{\underset{\uparrow}{1}, 1, 1\}$

(b) $x_1(n) = (\frac{1}{2})^n u(n)$, $\qquad x_2(n) = (\frac{1}{3})^n u(n)$

(c) $x_1(n) = \{1, \underset{\uparrow}{2}, 3, 4\}$, $\qquad x_2(n) = \{4, 3, \underset{\uparrow}{2}, 1\}$

(d) $x_1(n) = \{\underset{\uparrow}{1}, 1, 1, 1, 1\}$, $\qquad x_2(n) = \{\underset{\uparrow}{1}, 1, 1\}$

Did you obtain the same results by both methods? Explain.

3.32 Determine the one-sided z-transform of the constant signal $x(n) = 1$, $-\infty < n < \infty$.

3.33 Prove that the Fibonacci sequence can be thought of as the impulse response of the system described by the difference equation $y(n) = y(n-1) + y(n-2) + x(n)$. Then determine $h(n)$ using z-transform techniques.

3.34 Use the one-sided z-transform to determine $y(n)$, $n \ge 0$ in the following cases.

(a) $y(n) + \frac{1}{2}y(n-1) - \frac{1}{4}y(n-2) = 0$; $\quad y(-1) = y(-2) = 1$

(b) $y(n) - 1.5y(n-1) + 0.5y(n-2) = 0$; $\quad y(-1) = 1, y(-2) = 0$

(c) $y(n) = \frac{1}{2}y(n-1) + x(n)$

$x(n) = (\frac{1}{3})^n u(n)$, $\quad y(-1) = 1$

(d) $y(n) = \frac{1}{4}y(n-2) + x(n)$

$x(n) = u(n)$

$y(-1) = 0$; $\quad y(-2) = 1$

3.35 Show that the following systems are equivalent.

(a) $y(n) = 0.2y(n-1) + x(n) - 0.3x(n-1) + 0.02x(n-2)$

(b) $y(n) = x(n) - 0.1x(n-1)$

3.36 Consider the sequence $x(n) = a^n u(n)$, $-1 < a < 1$. Determine at least two sequences that are not equal to $x(n)$ but have the same autocorrelation.

3.37 Compute the unit step response of the system with impulse response

$$h(n) = \begin{cases} 3^n, & n < 0 \\ (\frac{2}{5})^n, & n \geq 0 \end{cases}$$

3.38 Compute the zero-state response for the following pairs of systems and input signals.

(a) $h(n) = (\frac{1}{3})^n u(n)$, $x(n) = (\frac{1}{2})^n \left(\cos \frac{\pi}{3} n\right) u(n)$

(b) $h(n) = (\frac{1}{4})^n u(n)$, $x(n) = (\frac{1}{3})^n u(n) + (\frac{1}{2})^{-n} u(-n-1)$

(c) $y(n) = -0.1y(n-1) + 0.2y(n-2) + x(n) + x(n-1)$
$x(n) = (\frac{1}{3})^n u(n)$

(d) $y(n) = \frac{1}{2}x(n) - \frac{1}{2}x(n-1)$
$x(n) = 10\left(\cos \frac{\pi}{2} n\right) u(n)$

(e) $y(n) = -y(n-2) + 10x(n)$
$x(n) = 10\left(\cos \frac{\pi}{2} n\right) u(n)$

(f) $h(n) = (\frac{2}{5})^n u(n)$, $x(n) = u(n) - u(n-7)$

(g) $h(n) = (\frac{1}{2})^n u(n)$, $x(n) = (-1)^n$, $-\infty < n < \infty$

(h) $h(n) = (\frac{1}{2})^n u(n)$, $x(n) = (n+1)(\frac{1}{4})^n u(n)$

3.39 Consider the system

$$H(z) = \frac{1 - 2z^{-1} + 2z^{-2} - z^{-3}}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.2z^{-1})} \qquad \text{ROC: } 0.5 < |z| < 1$$

(a) Sketch the pole–zero pattern. Is the system stable?
(b) Determine the impulse response of the system.

3.40 Compute the response of the system

$$y(n) = 0.7y(n-1) - 0.12y(n-2) + x(n-1) + x(n-2)$$

to the input $x(n) = nu(n)$. Is the system stable?

3.41 Determine the impulse response and the step response of the following causal systems. Plot the pole–zero patterns and determine which of the systems are stable.

(a) $y(n) = \frac{3}{4}y(n-1) - \frac{1}{8}y(n-2) + x(n)$

(b) $y(n) = y(n-1) - 0.5y(n-2) + x(n) + x(n-1)$

(c) $H(z) = \dfrac{z^{-1}(1 + z^{-1})}{(1 - z^{-1})^3}$

(d) $y(n) = 0.6y(n-1) - 0.08y(n-2) + x(n)$

(e) $y(n) = 0.7y(n-1) - 0.1y(n-2) + 2x(n) - x(n-2)$

3.42 Let $x(n)$ be a causal sequence with z-transform $X(z)$ whose pole–zero plot is shown in Fig. P3.42. Sketch the pole–zero plots and the ROC of the following sequences:

(a) $x_1(n) = x(-n+2)$

(b) $x_2(n) = e^{j(\pi/3)n} x(n)$

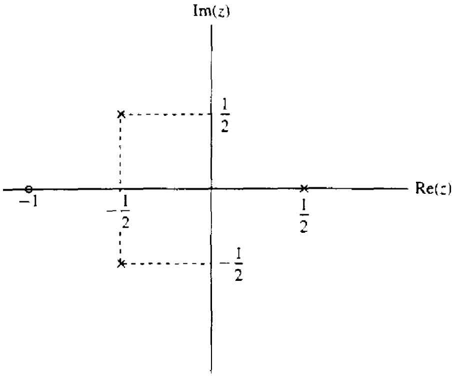


Figure P3.42

3.43 We want to design a causal discrete-time LTI system with the property that if the input is

$$x(n) = (\tfrac{1}{2})^n u(n) - \tfrac{1}{4}(\tfrac{1}{2})^{n-1} u(n-1)$$

then the output is

$$y(n) = (\tfrac{1}{3})^n u(n)$$

(a) Determine the impulse response $h(n)$ and the system function $H(z)$ of a system that satisfies the foregoing conditions.
(b) Find the difference equation that characterizes this system.
(c) Determine a realization of the system that requires the minimum possible amount of memory.
(d) Determine if the system is stable.

3.44 Determine the stability region for the causal system

$$H(z) = \frac{1}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

by computing its poles and restricting them to be inside the unit circle.

3.45 Consider the system

$$H(z) = \frac{z^{-1} + \frac{1}{2}z^{-2}}{1 - \frac{3}{5}z^{-1} + \frac{2}{25}z^{-2}}$$

Determine:
(a) The impulse response
(b) The zero-state step response
(c) The step response if $y(-1) = 1$ and $y(-2) = 2$

3.46 Determine the system function, impulse response, and zero-state step response of the system shown in Fig **P3.46.**

3.47 Consider the causal system

$$y(n) = -a_1 y(n-1) + b_0 x(n) + b_1 x(n-1)$$

Determine:
(a) The impulse response

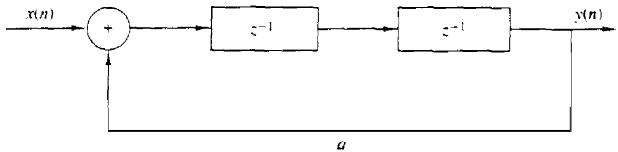


Figure P3.46

(b) The zero-state step response
(c) The step response if $y(-1) = A \neq 0$
(d) The response *to* the input

$$x(n) = \cos \omega_0 n \qquad 0 \leq n < \infty$$

3.48 Derermine the zero-state response of the system

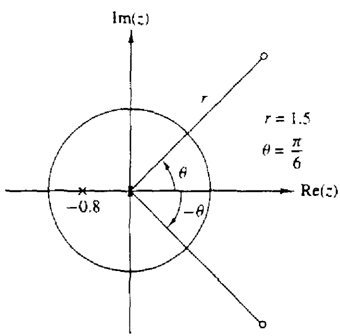
$$y(n) = \tfrac{1}{2}y(n-1) + 4x(n) + 3x(n-1)$$

to the input

$$x(n) = e^{j\omega_0 n}u(n)$$

What is the steady-state response of the system?

3.49 Consider the causal system defined by the pole–zero pattern shown in Fig. P3.49.
(a) Determine the system function and the impulse response of the system given that $H(z)|_{z=1} = 1$.
(h) Is the system stable?
(c) Sketch a possible implementation of the system and determine the corresponding difference equations,



F i r e P3.49

3.50 An FIR LTI system has an impulse response $h(n)$, which is real valued, even, and has finite **duration of** $2N + 1$. Show that if $z_1 = re^{j\omega_0}$ is a zero of the system, then $z_1 = (1/r)e^{j\omega_0}$ is **also** a zero.

351 Consider an LTI discrete-time system whose pole–zero pattern is shown **in** Fig. P3.51.
(a) Determine the ROC of the system function $H(z)$ if the system **is known** to be stable.

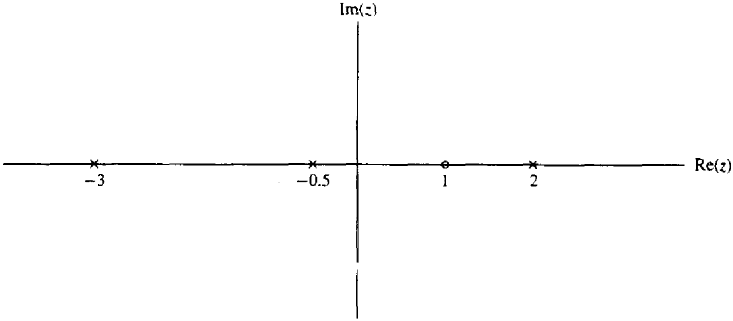


Figure P3.51

(b) It is possible for the given pole–zero plot to correspond to a causal and stable system? If so, what is the appropriate ROC?
(c) How many possible systems can be associated with this pole–zero pattern?

3.52 Let $x(n)$ be a causal sequence.
(a) What conclusion can you draw about the value of its z-transform $X(z)$ at $z = \infty$?
(b) Use the result in part (a) to check which of the following transforms cannot be associated with a causal sequence.

$$\text{(i)}\ X(z) = \frac{(z - \frac{1}{2})^4}{(z - \frac{1}{3})^3} \qquad \text{(ii)}\ X(z) = \frac{(1 - \frac{1}{2}z^{-1})^2}{(1 - \frac{1}{3}z^{-1})} \qquad \text{(iii)}\ X(z) = \frac{(z - \frac{1}{3})^2}{(z - \frac{1}{2})^3}$$

3.53 A causal pole–zero system is BIBO stable if its poles are inside the unit circle. Consider now a pole–zero system that is BIBO stable and has its poles inside the unit circle. Is the system always causal? [*Hint:* Consider the systems $h_1(n) = a^n u(n)$ and $h_2(n) = a^n u(n+3)$, $|a| < 1$.]

3.54 Let $x(n)$ be an anticausal signal [i.e., $x(n) = 0$ for $n > 0$]. Formulate and prove an initial value theorem for anticausal signals.

3.55 The step response of an LTI system is

$$s(n) = (\tfrac{1}{3})^{n-2} u(n+2)$$

(a) Find the system function $H(z)$ and sketch the pole–zero plot.
(b) Determine the impulse response $h(n)$.
(c) Check if the system is causal and stable.

3.56 Use contour integration to determine the sequence $x(n)$ whose z-transform is given by

(a) $X(z) = \dfrac{1}{1 - \frac{1}{2}z^{-1}} \qquad |z| > \frac{1}{2}$

(b) $X(z) = \dfrac{1}{1 - \frac{1}{2}z^{-1}} \qquad |z| < \frac{1}{2}$

(c) $X(z) = \dfrac{z-a}{1-az} \qquad |z| > |1/a|$

(d) $X(z) = \dfrac{1-\frac{1}{4}z^{-1}}{1-\frac{1}{6}z^{-1}-\frac{1}{6}z^{-2}} \qquad |z| > \frac{1}{2}$

3.57 Let $x(n)$ be a sequence with z-transform

$$X(z) = \frac{1-a^2}{(1-az)(1-az^{-1})} \qquad \text{ROC: } a < |z| < 1/a$$

with $0 < a < 1$. Determine $x(n)$ by using contour integration

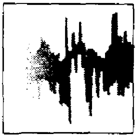
358 The z-transform of a sequence $x(n)$ is given by

$$X(z) = \frac{z^{20}}{(z-\frac{1}{2})(z-2)^5(z+\frac{5}{2})^2(z+3)}$$

Furthermore it is known that $X(z)$ converges for $|z| = 1$.

(a) Determine the ROC of $X(z)$.

(b) Determine $x(n)$ at $n = -18$. **(Hint:** Use contour integration.)



4

Frequency Analysis of Signals and Systems

The Fourier transform is one of several mathematical tools that is useful in the analysis and design of LTI systems. Another is the Fourier series. These signal representations basically involve the decomposition of the signals in terms of sinusoidal (or complex exponential) components. With such a decomposition, a signal is said to be represented in the *frequency* domain.

As we shall demonstrate, most signals of practical interest can be decomposed into a sum of sinusoidal signal components. For the class of periodic signals, such a decomposition is called a Fourier *series*. For the class of finite energy signals, the decomposition is called the *Fourier transform*. These decompositions are extremely important in the analysis of LTI systems because the response of an LTI system to a sinusoidal input signal is a sinusoid of the same frequency but of different amplitude and phase. Furthermore. the linearity property of the LTI system implies that a linear sum of sinusoidal components at the input produces a similar linear sum of sinusoidal components at the output, which differ only in the amplitudes and phases from the input sinusoids. This characteristic behavior of LTI systems renders the sinusoidal decomposition of signals very important. Although many other decompositions of signals are possible, only the class of sinusoidal (or complex exponential) signals possess this desirable property in passing through an LTI system.

We begin our study of frequency analysis of signals with the representation of continuous-time periodic and aperiodic signals by means of the Fourier series and the Fourier transform. respectively. This is followed by a parallel treatment of discrete-time periodic and aperiodic signals. The properties of the Fourier transform are described in detail and a number of time-frequency dualities are presented.

4.1 FREQUENCY ANALYSIS OF CONTINUOUS-TIME SIGNALS

It is well known that a prism can be used to break up white light (sunlight) into the colors of the rainbow (see Fig. 4.1a). In a paper submitted in 1672 to the Royal Society, Isaac Newton used the term ***spectrum*** to describe the ***continuous*** bands

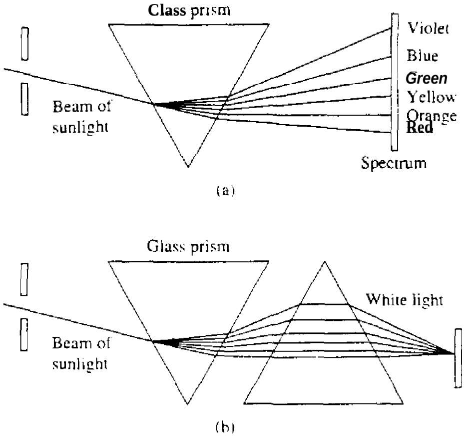


Figure 4.1 (a) Analysis and (b) synthesis of the white light (sunlight) using glass prisms.

of colors produced by this apparatus. To understand this phenomenon. Newton placed another prism upside-down with respect to the first. and showed that the colors blended hack into white light. as in Fig. 4.1b. By inserting a slit between the two prisms and blocking one or more colors from hitting the second prism. he showed that the remixed light is no longer white. Hence the light passing through the first prism is simply analyzed into its component colors without any other change. However. only if we mix again all of these colors do we obtain the original white light.

Later. Joseph Fraunhofer (1787–1826). in making measurements of light emitted by the sun and stars. discovered that the spectrum of the observed light consists of distinct color lines. A few years later (mid-1800s) Gustav Kirchhoff and Robert Bunsen found that each chemical element. when heated to incandescence. radiated its own distinct color of light. As a consequence. each chemical element can be identified by its own *line* spectrum.

From physics we know that each color corresponds to a specific frequency of the visible spectrum. Hence the analysis of light into colors is actually a form of frequency analysis.

Frequency analysis of a signal involves the resolution of the signal into its frequency (sinusoidal) components. Instead of light, our signal waveforms are basically functions of time. The role of the prism is played by the Fourier analysis tools that we will develop: the Fourier series and the Fourier transform. The recombination of the sinusoidal components to reconstruct the original signal is basically a Fourier synthesis problem. The problem of signal analysis is basically the same for the case of a signal waveform and for the case of the light from heated chemical compositions. Just as in the case of chemical compositions. different signal waveforms have different spectra. Thus the spectrum provides an "identity" or a signature for the signal in the sense that no other signal has the same spectrum. As we will see, this attribute is related to the mathematical treatment of frequency-domain techniques.

If we decompose a waveform into sinusoidal components. in much the same way that a prism separates white light into different colors, the sum of these sinusoidal components results in the original waveform. On the other hand, if any of these components is missing, the result is a different signal.

In our treatment of frequency analysis, we will develop the proper mathematical tools ("prisms") for the decomposition of signals ("light") into sinusoidal frequency components (colors). Furthermore, the tools ("inverse prisms") for synthesis of a given signal from its frequency components will also be developed.

The basic motivation for developing the frequency analysis tools is to provide a mathematical and pictorial representation for the frequency components that are contained in any given signal. As in physics, the term *spectrum* is used when referring to the frequency content of a signal. The process of obtaining the spectrum of a given signal using the basic mathematical tools described in this chapter is known as *frequency* or *spectral analysis.* In contrast, the process of determining the spectrum of a signal in practice, based on actual measurements of the signal, is called *spectrum estimation.* This distinction is very important. In a practical problem the signal to be analyzed does not lend itself to an exact mathematical description. The signal is usually some information-bearing signal from which we are attempting to extract the relevant information. If the information that we wish to extract can be obtained either directly or indirectly from the spectral content of the signal, we can perform *spectrum estimation* on the information-bearing signal, and thus obtain an estimate of the signal spectrum. In fact, we can view spectral estimation as a type of spectral analysis performed on signals obtained from physical sources (e.g., speech, EEG, ECG, etc.). The instruments or software programs used to obtain spectral estimates of such signals are known as *spectrum analyzers.*

Here, we will deal with spectral analysis. However, in Chapter 12 we shall treat the subject of power spectrum estimation.

4.1.1 The Fourier Series for Continuous-Time Periodic Signals

In this section we present the frequency analysis tools for continuous-time periodic signals. Examples of periodic signals encountered in practice are square waves, rectangular waves, triangular waves, and of course, sinusoids and complex exponentials.

The basic mathematical representation of periodic signals is the Fourier series, which is a linear weighted sum of harmonically related sinusoids or complex exponentials. Jean Baptiste Joseph Fourier (1768–1830), a French mathematician, used such trigonometric series expansions in describing the phenomenon of heat conduction and temperature distribution through bodies. Although his work was motivated by the problem of heat conduction, the mathematical techniques that he developed during the early part of the nineteenth century now find application in a variety of problems encompassing many different fields, including optics, vibrations in mechanical systems, system theory, and electromagnetics.

From Chapter 1 we recall that a linear combination of harmonically related complex exponentials of the form

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k F_0 t} \tag{4.1.1}$$

is a periodic signal with fundamental period $T_p = 1/F_0$. Hence we can think of the exponential signals

$$\{e^{j2\pi k F_0 t} \qquad k = 0, \pm 1, \pm 2, \ldots\}$$

as the basic "building blocks" from which we can construct periodic signals of various types by proper choice of the fundamental frequency and the coefficients $\{c_k\}$. F_0 determines the fundamental period of x(t) and the coefficients $\{c_k\}$ specify the shape of the waveform.

Suppose that we are given a periodic signal $x(t)$ with period T_p. We can represent the periodic signal by the series (4.1.1), called a *Fourier series,* where the fundamental frequency F_0 is selected to be the reciprocal of the given period T_p. To determine the expression for the coefficients $\{c_k\}$, we first multiply both sides of (4.1.1) by the complex exponential

$$e^{-j2\pi F_0 l t}$$

where l is an integer and then integrate both sides of the resulting equation over a single period, say from 0 to T_p, or more generally, from t_0 to $t_0 + T_p$, where t_0 is an arbitrary but mathematically convenient starting value. Thus we obtain

$$\int_{t_0}^{t_0+T_p} x(t)e^{-j2\pi l F_0 t}\,dt = \int_{t_0}^{t_0+T_p} e^{-j2\pi l F_0 t} \left(\sum_{k=-\infty}^{\infty} c_k e^{+j2\pi k F_0 t}\right) dt \tag{4.1.2}$$

To evaluate the integral on the righi-hand side of (4.1.2), we interchange the order of the summation and integration and combine the two exponentials. Hence

$$\sum_{k=-\infty}^{\infty} c_k \int_{t_0}^{t_0+T_p} e^{j2\pi F_0(k-l)t}\,dt = \sum_{k=-\infty}^{\infty} c_k \left[\frac{e^{j2\pi F_0(k-l)t}}{j2\pi F_0(k-l)}\right]_{t_0}^{t_0+T_p} \tag{4.1.3}$$

For $k \neq l$, the right-hand side of (4.1.3) evaluated at the lower and upper limits, t_0 and $t_0 + T_p$, respectively, yields zero. On the other hand, if $k = l$, we have

$$\int_{t_0}^{t_0+T_p} dt = t\Big]_{t_0}^{t_0+T_p} = T_p$$

Consequently, (4.1.2) reduces to

$$\int_{t_0}^{t_0+T_p} x(t)e^{-j2\pi l F_0 t}\,dt = c_l T_p$$

and therefore the expression for the Fourier coefficients in terms of the given periodic signal becomes

$$c_l = \frac{1}{T_p} \int_{t_0}^{t_0+T_p} x(t) e^{-j2\pi l F_0 t} dt$$

Since t_0 is arbitrary, this integral can be evaluated over any interval of length T_p, that is, over any interval equal to the period of the signal $x(t)$. Consequently, the integral for the Fourier series coefficients will be written as

$$c_l = \frac{1}{T_p} \int_{T_p} x(t) e^{-j2\pi l F_0 t} dt \tag{4.1.4}$$

An important issue that arises in the representation of the periodic signal $x(t)$ by the Fourier series is whether or not the series converges to $x(t)$ for every value of t, that is, if the signal $x(t)$ and its Fourier series representation

$$\sum_{k=-\infty}^{\infty} c_k e^{j2\pi k F_0 t} \tag{4.1.5}$$

are equal at every value of t. The so-called *Dirichlet conditions* guarantee that the series (4.1.5) will be equal to $x(t)$, except at the values of t for which $x(t)$ is discontinuous. At these values of t, (4.1.5) converges to the midpoint (average value) of the discontinuity. The Dirichlet conditions are:

1. The signal $x(t)$ has a finite number of discontinuities in any period.
2. The signal $x(t)$ contains a finite number of maxima and minima during any period.
3. The signal $x(t)$ is absolutely integrable in any period, that is.

$$\int_{T_p} |x(t)| dt < \infty \tag{4.1.6}$$

All periodic signals of practical interest satisfy these conditions.

The weaker condition, that the signal has finite energy in one period.

$$\int_{T_p} |x(t)|^2 dt < \infty \tag{4.1.7}$$

guarantees that the energy in the difference signal

$$e(t) = x(t) - \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k F_0 t}$$

is zero, although $x(t)$ and its Fourier series may not be equal for all values of t. Note that (4.1.6) implies (4.1.7), but not vice versa. Also, both (4.1.7) and the Dirichlet conditions are sufficient but not necessary conditions (i.e., there are signals that have a Fourier series representation but do not satisfy these conditions).

In summary, if $x(t)$ is periodic and satisfies the Dirichlet conditions, it can be represented in a Fourier series as in (4.1.1), where the coefficients are specified by (4.1.4). These relations are summarized below.

FREQUENCY ANALYSIS OF CONTINUOUS-TIME PERIODIC SIGNALS

Synthesis equation	$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k F_0 t}$	(4.1.8)
Analysis equation	$c_k = \frac{1}{T_p} \int_{T_p} x(t) e^{-j2\pi k F_0 t} dt$	(4.1.9)

In general, the Fourier coefficients c_k are complex valued. Moreover, it is easily shown that if the periodic signal is real, c_k and c_{-k} are complex conjugates. As a result, if

$$c_k = |c_k| e^{j\theta_k}$$

then

$$c_{-k} = |c_k|^{-j\theta_k}$$

Consequently, the Fourier series may also be represented in the form

$$x(t) = c_0 + 2 \sum_{k=1}^{\infty} |c_k| \cos(2\pi k F_0 t + \theta_k) \tag{4.1.10}$$

where c_0 is real valued when $x(t)$ is real.

Finally, we should indicate that yet another form for the Fourier series can be obtained by expanding the cosine function in (4.1.10) as

$$\cos(2\pi k F_0 t + \theta_k) = \cos 2\pi k F_0 t \cos\theta_k - \sin 2\pi k F_0 t \sin\theta_k$$

Consequently, we can rewrite (4.1.10) in the form

$$x(t) = a_0 + \sum_{k=1}^{\infty} (a_k \cos 2\pi k F_0 t - b_k \sin 2\pi k F_0 t) \tag{4.1.11}$$

where

$$\begin{aligned} a_0 &= c_0 \\ a_k &= 2|c_k| \cos\theta_k \\ b_k &= 2|c_k| \sin\theta_k \end{aligned}$$

The expressions in (4.1.8), (4.1.10), and (4.1.11) constitute three equivalent forms for the Fourier series representation of a real periodic signal.

4.1.2 Power Density Spectrum of Periodic Signals

A periodic signal has infinite energy and a finite average power, which is given as

$$P_x = \frac{1}{T_p} \int_{T_p} |x(t)|^2 dt \tag{4.1.12}$$

If we take the complex conjugate of (4.1.8) and substitute for $x^*(t)$ in (4.1.12), we obtain

$$\begin{aligned} P_x &= \frac{1}{T_p}\int_{T_p} x(t) \sum_{k=-\infty}^{\infty} c_k^* e^{-j2\pi kF_0 t}\,dt \\ &= \sum_{k=-\infty}^{\infty} c_k^* \left[\frac{1}{T_p}\int_{T_p} x(t) e^{-j2\pi kF_0 t}\,dt\right] \\ &= \sum_{k=-\infty}^{\infty} |c_k|^2 \end{aligned} \tag{4.1.13}$$

Therefore, we have established the relation

$$P_x = \frac{1}{T_p}\int_{T_p} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2 \tag{4.1.14}$$

which is called *Parseval's relation* for power signals.

To illustrate the physical meaning of (4.1.14), suppose that $x(t)$ consists of a single complex exponential

$$x(t) = c_k e^{j2\pi kF_0 t}$$

In this case, all the Fourier series coefficients except c_k are zero. Consequently, the average power in the signal is

$$P_x = |c_k|^2$$

It is obvious that $|c_k|^2$ represents the power in the kth harmonic component of the signal. Hence the total average power in the periodic signal is simply the sum of the average powers in all the harmonics.

If we plot the $|c_k|^2$ as a function of the frequencies kF_0, $k = 0, \pm1, \pm2, \ldots$, the diagram that we obtain shows how the power of the periodic signal is distributed among the various frequency components. This diagram, which is illustrated in Fig. 4.2, is called the *power density spectrum** of the periodic signal $x(t)$. Since the

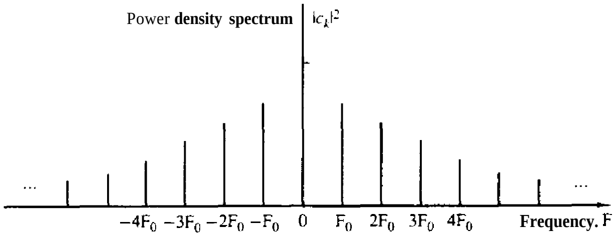


Figure 4.2 **Power density spectrum of a continuous-time periodic signal.**

*This function is also called the *power spectral density* or, simply, the *power spectrum.*

power in a periodic signal exists only at discrete values of frequencies (i.e., F = 0. $\pm F_0$. $\pm 2F_0, \ldots$). the signal is said to have a *line spectrum*. The spacing between two consecutive spectral lines is equal to the reciprocal of the fundamental period T_p, whereas the shape of the spectrum (i.e.. the power distribution of the signal). depends on the time-domain characteristics of the signal.

As indicated in the preceding section, the Fourier series coefficients $\{c_k\}$ are complex valued. that is. they can be represented as

$$c_k = |c_k|e^{j\theta_k}$$

where

$$\theta_k = \measuredangle c_k$$

Instead of plotting the power density spectrum. we can plot the magnitude voltage spectrum $\{|c_k|\}$ and the phase spectrum $\{\theta_k\}$ as a function of frequency. Clearly. the power spectral density in the periodic signal is simply the square of the magnitude spectrum. The phase information is totally destroyed (or does not appear) in the power spectral density.

If the periodic signal is real valued. the Fourier series coefficients $\{c_k\}$ satisfy the condition

$$c_{-k} = c_k^*$$

Consequently. $|c_k|^2 = |c_k^*|^2$. Hence the power spectrum is a symmetric function of frequency. This condition also means that the magnitude spectrum is svmmetric (even function) about the origin and the phase spectrum is an odd function. As a consequence of the symmetry. it is sufficient to specify the spectrum of a real periodic signal for positive frequencies only. Furthermore. the total average power can be expressed as

$$P_x = c_0^2 + 2\sum_{k=1}^{\infty} |c_k|^2 \tag{4.1.15}$$

$$= a_0^2 + \frac{1}{2}\sum_{k=1}^{\infty}(a_k^2 + b_k^2) \tag{4.1.16}$$

which follows directly from the relationships given in Section 4.1.1 among $\{a_k\}$, $\{b_k\}$, and $\{c_k\}$ coefficients in the Fourier series expressions.

Example 4.1.1

Determine the Fourier series and the power density spectrum of the rectangular pulse train signal illustrated in Fig. 4.3.

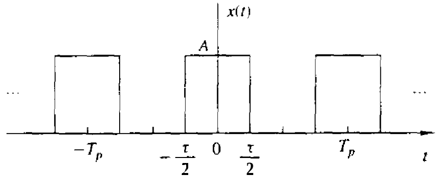


Figure 4.3 Continuous-time periodic train of rectangular pulses.

Solution The signal is periodic with fundamental period T_p and, clearly, satisfies the Dirichlet conditions. Consequently, we can represent the signal in the Fourier series given by (4.1.8) with the Fourier coefficients specified by (4.1.9).

Since $x(t)$ is an even signal [i.e., $x(t) = x(-t)$], it is convenient to select the integration interval from $-T_p/2$ to $T_p/2$. Thus (4.1.9) evaluated for $k = 0$ yields

$$c_0 = \frac{1}{T_p}\int_{-T_p/2}^{T_p/2} x(t)dt = \frac{1}{T_p}\int_{-\tau/2}^{\tau/2} A dt = \frac{A\tau}{T_p} \tag{4.1.17}$$

The term c_0 represents the average value (dc component) of the signal $x(t)$. For $k \neq 0$ we have

$$\begin{aligned} c_k &= \frac{1}{T_p}\int_{-\tau/2}^{\tau/2} A e^{-j2\pi k F_0 t} dt = \frac{A}{T_p}\left[\frac{e^{-j2\pi F_0 k t}}{-j2\pi k F_0}\right]_{-\tau/2}^{\tau/2} \\ &= \frac{A}{\pi F_0 k T_p}\frac{e^{j\pi k F_0 \tau} - e^{-j\pi k F_0 \tau}}{j2} \\ &= \frac{A\tau}{T_p}\frac{\sin \pi k F_0 \tau}{\pi k F_0 \tau} \qquad k = \pm 1, \pm 2, \ldots \end{aligned} \tag{4.1.18}$$

It is interesting to note that the right-hand side of (4.1.18) has the form $(\sin\phi)/\phi$, where $\phi = \pi k F_0 \tau$. In this case ϕ takes on discrete values since F_0 and τ are fixed and the index k varies. However, if we plot $(\sin\phi)/\phi$ with ϕ as a continuous parameter over the range $-\infty < \phi < \infty$, we obtain the graph shown in Fig. 4.4. We observe that this function decays to zero as $\phi \to \pm\infty$, has a maximum value of unity at $\phi = 0$, and is zero at multiples of π (i.e., at $\phi = m\pi$, $m = \pm 1, \pm 2, \ldots$). It is clear that the Fourier coefficients given by (4.1.18) are the sample values of the $(\sin\phi)/\phi$ function for $\phi = \pi k F_0 \tau$ and scaled in amptitude by $A\tau/T_p$.

Since the periodic function $x(t)$ is even, the Fourier coefficients c_k are real. Consequently, the phase spectrum is either zero, when c_k is positive, or π when c_k is negative. Instead of plotting the magnitude and phase spectra separately, we may simply plot $\{c_k\}$ on a single graph, showing both the positive and negative values c_k on the graph. This is commonly done in practice when the Fourier coefficients $\{c_k\}$ are real.

Figure 4.5 illustrates the Fourier coefficients of the rectangular pulse train when T_p is fixed and the pulse width τ is allowed to vary. In this case $T_p = 0.25$ second, so that $F_0 = 1/T_p = 4$ Hz and $\tau = 0.05T_p$, $\tau = 0.1T_p$, and $\tau = 0.2T_p$. We observe that the effect of decreasing τ while keeping T_p fixed is to spread out the signal power over the frequency range. The spacing between adjacent spectral lines is $F_0 = 4$ Hz, independent of the value of the pulse width τ.

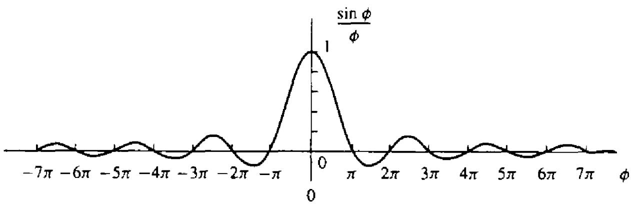


Figure 4.4 The function $(\sin\phi)/\phi$.

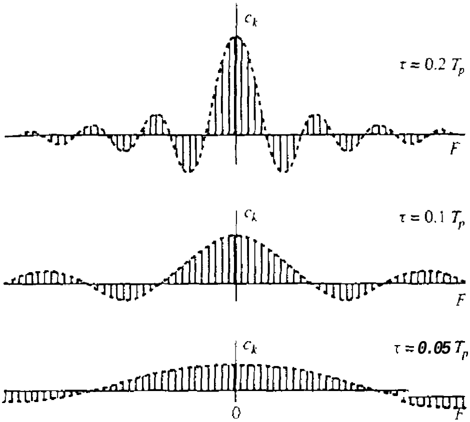


Figure 4.5 Fourier coefficients of the rectangular pulse train with T_p is **fixed** and the pulse width τ **varies.**

On the other hand. it is also instructive to fix τ and vary the period T_p when $T_p > \tau$. Figure 4.6 illustrates this condition when $T_p = 5\tau$, $T_p = 10\tau$, and $T_p = 20\tau$. In this case. the spacing between adjacent spectral lines decreases as T„ increases. In the limit as $T_p \to \infty$, the Fourier coefficients c_k approach zero due to the factor of T„ in the denominator of (4.1.18). This behavior is consistent with the fact that as $T_p \to \infty$ and τ remains fixed, the resulting signal is no longer a power signal. Instead.

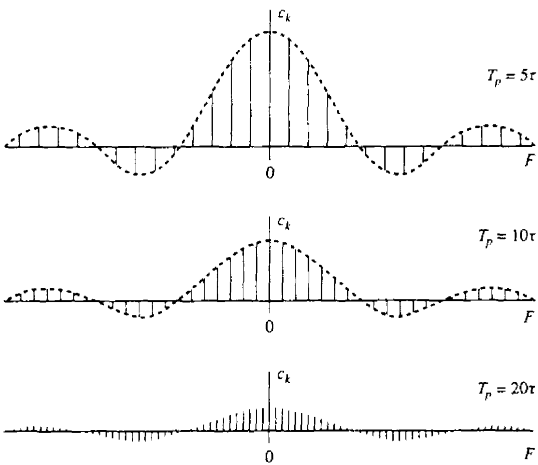


Figure 4.6 Fourier coefficient of a rectangular pulse train with **fixed** pulse width τ and varying period T_p.

it becomes an energy signal and its average power is zero. The spectra of finite energy signals are described in the next section.

We also note that if $k \# 0$ **and** $\sin(\pi k F_0 \tau) = 0$, **then** $c_k = 0$. **The harmonics with zero power occur at frequencies** kF_0 **such that** $\pi(kF_0)\tau = m\pi$, $m = \pm 1, \pm 2, \ldots$, **or at** $kF_0 = m/\tau$. **For example, if** $F_0 = 4$ **Hz and** $\tau = 0.27$,**. it follows that the spectral components at ±20 Hz, 140** $Hz, \ldots$ **have zero power. These frequencies correspond to the Fourier coefficients** c_k, $k = f5, 110, 115, \ldots$. **On the other hand. if** $\tau = 0.1T_p$, **the spectral components with zero power are** $k = \pm 10, \pm 20, \pm 30, \ldots$.

The power density spectrum far the rectangular pulse train is

$$|c_k|^2 = \begin{cases} \left(\dfrac{A\tau}{T_p}\right)^2, & k = 0 \\ \left(\dfrac{A\tau}{T_p}\right)^2 \left(\dfrac{\sin \pi k F_0 \tau}{\pi k F_0 \tau}\right)^2, & k = \pm 1, \pm 2, \ldots \end{cases} \tag{4.1.19}$$

4.1.3 The Fourier Transform for Continuous-Time Aperiodic Signals

In Section 4.1.1 we developed the Fourier series to represent a periodic signal as a linear combination of harmonically related complex exponentials. As a consequence of the periodicity, we saw that these signals possess line spectra with equidistant lines. The line spacing is equal to the fundamental frequency, which in turn is the inverse of the fundamental period of the signal. We can view the fundamental period as providing the number of lines per unit of frequency (line density). as illustrated in Fig. 4.6.

With this interpretation in mind, it is apparent that if we allow the period to increase without limit, the line spacing tends toward zero. In the limit, when the period becomes infinite, the signal becomes aperiodic and its spectrum becomes continuous. This argument suggests that the spectrum of an aperiodic signal will be the envelope of the line spectrum in the corresponding periodic signal obtained by repeating the aperiodic signal with some period T_p.

Let us consider an aperiodic signal $x(t)$ with finite duration as shown in Fig. 4.7a. From this aperiodic signal. we can create a periodic signal $x_p(t)$ with period T_p, as shown in Fig. 4.7b. Clearly. $x_p(t) = x(t)$ in the limit as $T_p \to \infty$, that is,

$$x(t) = \lim_{T_p \to \infty} x_p(t)$$

This interpretation implies that we should be able to obtain the spectrum of $x(t)$ from the spectrum of $x_p(t)$ simply by taking the limit as $T_p \to \infty$.

We begin with the Fourier series representation of $x_p(t)$,

$$x_p(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k F_0 t} \qquad F_0 = \frac{1}{T_p} \tag{4.1.20}$$

where

$$c_k = \frac{1}{T_p} \int_{-T_p/2}^{T_p/2} x_p(t) e^{-j2\pi k F_0 t} \, dt \tag{4.1.21}$$

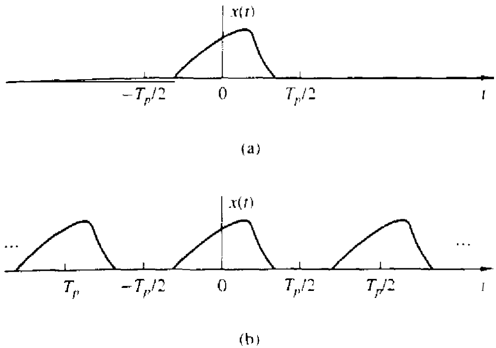


Figure 4.7 (a) Aperiodic signal $x(t)$ and (b) periodic signal $x_p(t)$ constructed by repeatinp $x(t)$ with a period T_p.

Since $x_p(t) = x(t)$ for $-T_p/2 \le t \le T_p/2$, (4.1.21) can be expressed as

$$c_k = \frac{1}{T_p}\int_{-T_p/2}^{T_p/2} x(t)e^{-j2\pi kF_0 t}\,dt \tag{4.1.22}$$

It is also true that $x(t) = 0$ for $|t| > T_p/2$. Consequently, the limits on the integral in (4.1.22) can be replaced by $-\infty$ and ∞. Hence

$$c_k = \frac{1}{T_p}\int_{-\infty}^{\infty} x(t)e^{-j2\pi kF_0 t}\,dt \tag{4.1.23}$$

Let us now define a function $X(F)$, called the Fourier transform of $x(t)$, as

$$X(F) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi Ft}\,dt \tag{4.1.24}$$

$X(F)$ is a function of the continuous variable F. It does not depend on T_p or F_0. However, if we compare (4.1.23) and (4.1.24). it is clear that the Fourier coefficients c_k can be expressed in terms of $X(F)$ as

$$c_k = \frac{1}{T_p}X(kF_0)$$

or equivalently.

$$T_p c_k = X(kF_0) = X\left(\frac{k}{T_p}\right) \tag{4.1.25}$$

Thus the Fourier coefficients are samples of $X(F)$ taken at multiples of F_0 and scaled by F_0 (multiplied by $1/T_p$). Substitution for c_k from (4.1.25) into (4.1.20) yields

$$x_p(t) = \frac{1}{T_p}\sum_{k=-\infty}^{\infty} X\left(\frac{k}{T_p}\right) e^{j2\pi kF_0 t} \tag{4.1.26}$$

We wish to take the limit of (4.1.26) as T_p approaches infinity. First. we define $A\text{F} = 1/T_p$. With this substitution. (4.1.26) becomes

$$x_p(t) = \sum_{k=-\infty}^{\infty} X(k\Delta F)e^{j2\pi k\Delta Ft}\Delta F \tag{4.1.27}$$

It is clear that in the limit as ***T,,*** approaches infinity, $x_p(t)$ reduces to $x(t)$. Also, ***AF*** becomes the differential dF and k***AF*** becomes the continuous frequency variable F. In turn. the summation in (4.1.27) becomes an integral over the frequency variable F. Thus

$$\begin{aligned} \lim_{T_p\to\infty} x_p(t) &= x(t) = \lim_{\Delta F\to 0} \sum_{k=-\infty}^{\infty} X(k\Delta F)e^{j2\pi k\Delta Ft} AF \\ x(t) &= \int_{-\infty}^{\infty} X(F)e^{j2\pi Ft}dF \end{aligned} \tag{4.1.28}$$

This integral relationship yields $x(t)$ when $X(F)$ is known, and it is called the in *erse* Fourier *transform*.

This concludes our heuristic derivation of the Fourier transform pair given by (4.1.24) and ***(4.1.28)*** for an aperiodic signal $x(t)$. Although the derivation is not mathematically rigorous, it led to the desired Fourier transform relationships with relatively simple intuitive arguments. In summary, the frequency analysis of continuous-time aperiodic signals involves the following Fourier transform pair.

FREQUENCY ANALYSIS Of CONTINUOUS-TIME APERIODIC SIGNALS

Synthesis equation inverse transform	$x(t) = \int_{-\infty}^{\infty} X(F)e^{j2\pi Ft}dF$	(4.1.29)
Analysis equation direct transform	$X(F) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi Ft}dt$	(4.1.30)

It is apparent that the essential difference between the Fourier series and the Fourier transform is that the spectrum in the latter case is continuous and hence the synthesis of an aperiodic signal from its spectrum is accomplished by means of integration instead of summation.

Finally. we wish to indicate that the Fourier transform pair in (4.1.29) and (4.1.30) can be expressed in terms of the radian frequency variable $\Omega = 2\pi F$. **Since** $dF = d\Omega/2\pi$. (4.1.29) and (4.1.30) become

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Omega)e^{j\Omega t}d\Omega \tag{4.1.31}$$

$$X(\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}dt \tag{4.1.32}$$

The set of conditions that guarantee the existence of the Fourier transform is the

Dirichlet conditions, which may be expressed as:

1. The signal $x(t)$ has a finite number of finite discontinuities.
2. The signal $x(t)$ has a finite number of maxima and minima.
3. The signal $x(t)$ is absolutely integrable, that is.

$$\int_{-\infty}^{\infty} |x(t)|dt < \infty \tag{4.1.33}$$

The third condition follows easily from the definition of the Fourier transform, given in (4.1.30). Indeed.

$$|X(F)| = \left| \int_{-\infty}^{\infty} x(t)e^{-j2\pi Ft}dt \right| \leq \int_{-\infty}^{\infty} |x(t)|dt$$

Hence $|X(F)| < \infty$ if (4.1.33) is satisfied.

A weaker condition for the existence of the Fourier transform is that $x(t)$ has finite energy: that is.

$$\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty \tag{4.1.34}$$

Note that if a signal $x(t)$ is absolutely integrable. it will also have finite energy. That is. if

$$\int_{-\infty}^{\infty} |x(t)|dt < \infty$$

then

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt < \infty \tag{4.1.35}$$

However. the converse is not true. That is. a signal may have finite energy but may not be absolutely integrable. For example, the signal

$$x(t) = \frac{\sin 2\pi t}{\pi t} \tag{4.1.36}$$

is square integrable but is not absolutely integrable. This signal has the Fourier transform

$$X(F) = \begin{cases} 1, & |F| \leq 1 \\ 0, & |F| > 1 \end{cases} \tag{4.1.37}$$

Since this signal violates **(4.1.33).** it is apparent that the Dirichlet conditions are sufficient but not necessary for the existence of the Fourier transform. In any case. nearly all finite energy signals have a Fourier transform. so that we need not worry about the pathological signals, which are seldom encountered in practice.

4.1.4 Energy Density Spectrum of Aperiodic Signals

Let $x(t)$ be any finite energy signal with Fourier transform $X(F)$. Its energy is

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

which, in turn, may be expressed in terms of $X(F)$ as follows:

$$\begin{aligned} E_x &= \int_{-\infty}^{\infty} x(t)x^*(t)dt \\ &= \int_{-\infty}^{\infty} x(t)dt \left[\int_{-\infty}^{\infty} X^*(F)e^{-j2\pi Ft}dF\right] \\ &= \int_{-\infty}^{\infty} X^*(F)dF \left[\int_{-\infty}^{\infty} x(t)e^{-j2\pi Ft}dt\right] \\ &= \int_{-\infty}^{\infty} |X(F)|^2 dF \end{aligned}$$

Therefore, we conclude that

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(F)|^2 dF \tag{4.1.38}$$

This is ***Parseval's relation*** for aperiodic, finite energy signals and expresses the principle of conservation of energy in the time and frequency domains.

The spectrum $X(F)$ of a signal is in general. complex valued. Consequently, it is usually expressed in polar forms as

$$X(F) = |X(F)|e^{j\Theta(F)}$$

where $|X(F)|$ is the magnitude spectrum and $\Theta(F)$ is the phase spectrum,

$$\Theta(F) = \measuredangle X(F)$$

On the other hand, the quantity

$$S_{xx}(F) = |X(F)|^2 \tag{4.1.39}$$

which is the integrand in (4.1.38), represents the distribution of energy in the signal as a function of frequency. Hence $S_{xx}(F)$ is called the ***energy density spectrum*** of $x(t)$. ***The*** integral of $S_{xx}(F)$ over all frequencies gives the total energy in the signal. Viewed in another way, the energy in the signal $x(t)$ over a band of frequencies $F_1 \leq F \leq F_1 + \Delta F$ is

$$\int_{F_1}^{F_1+\Delta F} S_{xx}(F)dF$$

From **(4.1.39)** we observe that $S_{xx}(F)$ does not contain any phase information [i.e., $S_{xx}(F)$ is purely real and nonnegative]. Since the phase spectrum of $x(t)$ is not contained in $S_{xx}(F)$, it is impossible to reconstruct the signal given $S_{xx}(F)$.

Finally, as in the case of Fourier series, it is easily shown that if the signal $x(t)$ is real. then

$$|X(-F)| = |X(F)| \tag{4.1.40}$$

$$\measuredangle X(-F) = -\measuredangle X(F) \tag{4.1.41}$$

By combining (4.1.40) and (4.1.39), we obtain

$$S_{xx}(-F) = S_{xx}(F) \tag{4.1.42}$$

In other words, the energy density spectrum of a real signal has even symmetry.

Example 4.1.2

Determine the Fourier transform and the energy density spectrum of a rectangular pulse signal defined as

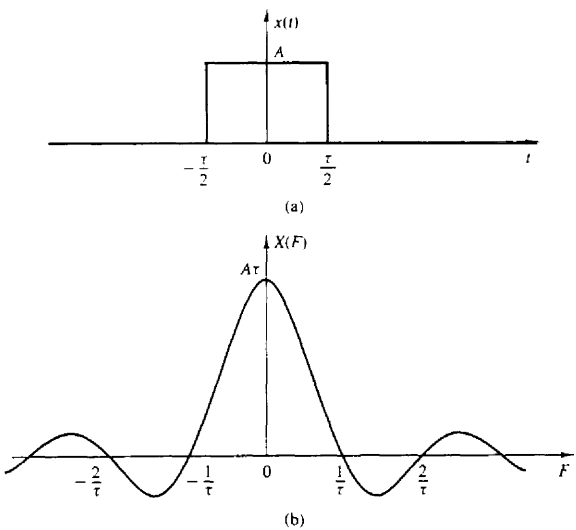
$$x(t) = \begin{cases} A, & |t| \le \tau/2 \\ 0, & |t| > \tau/2 \end{cases} \tag{4.1.43}$$

and illustrated in Fig. 4.8(a).

Solution Clearly, this signal is aperiodic and satisfies the Dirichlet conditions. Hence its Fourier transform exists. By applying (4.1.30), we find that

$$X(F) = \int_{-\tau/2}^{\tau/2} Ae^{-j2\pi Ft}\,dt = A\tau\frac{\sin \pi F\tau}{\pi F\tau} \tag{4.1.44}$$

We observe that $X(F)$ is real and hence it can be depicted graphically using only one diagram, as shown in Fig. 4.8(b). Obviously, $X(F)$ has the shape of the $(\sin\phi)/\phi$ function shown in Fig. 4.4. Hence the spectrum of the rectangular pulse is the envelope of the line spectrum (Fourier coefficients) of the periodic signal obtained by



Fire 4.8 (a) Rectangular pulse and (b) its Fourier transfonn.

periodically repeating the pulse with period T_p as in Fig. 4.3. In other words, the Fourier coefficients c_k in the corresponding periodic signal $x_p(t)$ are simply samples of $X(F)$ at frequencies $kF_0 = k/T_p$. Specifically,

$$c_k = \frac{1}{T_p} X(kF_0) = \frac{1}{T_p} X\left(\frac{k}{T_p}\right) \tag{4.1.45}$$

From (4.1.44) we note that the zero crossings of $X(F)$ occur at multiples of $1/\tau$. Furthermore, the width of the main lobe, which contains most of the signal energy, is equal to $2/\tau$. As the pulse duration τ decreases (increases), the main lobe becomes broader (narrower) and more energy is moved to the higher (lower) frequencies, as illustrated in Fig. 4.9. Thus as the signal pulse is expanded (compressed) in time, its transform is compressed (expanded) in frequency. This behavior. between the time function and its spectrum, is a type of uncertainty principle that appears in different forms in various branches of science and engineering.

Finally, the energy density spectrum of the rectangular pulse is

$$S_{xx}(F) = (A\tau)^2 \left(\frac{\sin \pi F\tau}{\pi F\tau}\right)^2 \tag{4.1.46}$$

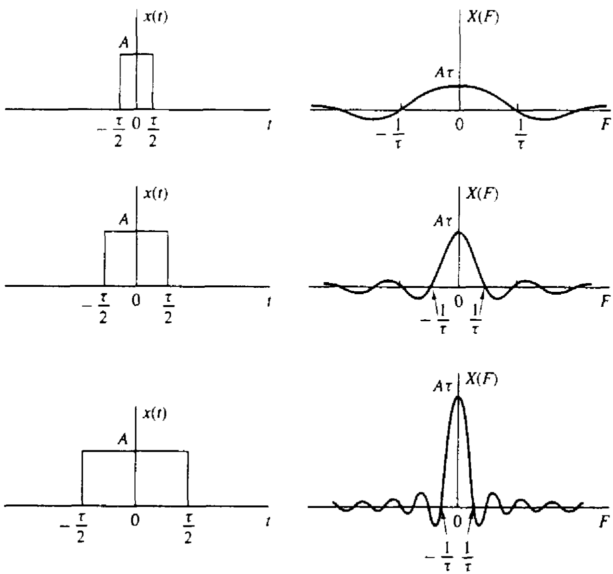


Figure 4.9 Fourier transform of a rectangular pulse for various width values.

4.2 FREQUENCY ANALYSIS OF DISCRETE-TIME SIGNALS

In Section 4.1 we developed the Fourier series representation for continuous-time periodic (power) signals and the Fourier transform for finite energy aperiodic signals. In this section we repeat the development for the class of discrete-time signals.

As we have observed from the discussion of Section 4.1, the Fourier series representation of a continuous-time periodic signal can consist of an infinite number of frequency components, where the frequency spacing between two successive harmonically related frequencies is $1/T_p$, and where T_p is the fundamental period. Since the frequency range for continuous-time signals extends from $-\infty$ to ∞, it is possible to have signals that contain an infinite number of frequency components. In contrast, the frequency range for discrete-time signals is unique over the interval $(-\pi, \pi)$ or $(0, 2\pi)$. A discrete-time signal of fundamental period **N** can consist of frequency components separated by $2\pi/N$ radians or $f = 1/N$ cycles. Consequently, the Fourier series representation of the discrete-time periodic signal will contain at most N frequency components. This is the basic difference between the Fourier series representations for continuous-time and discrete-time periodic signals.

4.2.1 The Fourier Series for Discrete-Time Periodic Signals

Suppose that we are given a periodic sequence $x(n)$ with period N, that is, $x(n) = x(n + N)$ for all n. The Fourier series representation for $x(n)$ consists of N harmonically related exponential functions

$$e^{j2\pi kn/N} \qquad k = 0, 1, \ldots, N-1$$

and is expressed as

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N} \tag{4.2.1}$$

where the $\{c_k\}$ are the coefficients in the series representation.

To derive the expression for the Fourier coefficients, we use the following formula:

$$\sum_{n=0}^{N-1} e^{j2\pi kn/N} = \begin{cases} N, & k = 0, \pm N, \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases} \tag{4.2.2}$$

Note the similarity of (4.2.2) with the continuous-time counterpart in (4.1.3). The proof of (4.2.2) follows immediately from the application of the geometric summation formula

$$\sum_{n=0}^{N-1} a^n = \begin{cases} N, & a = 1 \\ \dfrac{1-a^N}{1-a}, & a \neq 1 \end{cases} \tag{4.2.3}$$

The expression for the Fourier coefficients c_k can be obtained by multiplying both sides of **(4.2.1)** by the exponential $e^{-j2\pi ln/N}$ and summing the product from $n = 0$ to $n = N - 1$. Thus

$$\sum_{n=0}^{N-1} x(n)e^{-j2\pi ln/N} = \sum_{n=0}^{N-1}\sum_{k=0}^{N-1} c_k e^{j2\pi(k-l)n/N} \tag{4.2.4}$$

If we perform the summation over n first, in the right-hand side of (4.2.4), we obtain

$$\sum_{n=0}^{N-1} e^{j2\pi(k-l)n/N} = \begin{cases} N, & k - l = 0, \pm N, \pm 2N, \\ 0, & \text{otherwise} \end{cases} \tag{4.2.5}$$

where we have made use of (4.2.2). Therefore. the right-hand side of **(4.2.4)** reduces to Nc_l and hence

$$c_l = \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi ln/N} \qquad l = 0, 1, \ldots, N-1 \tag{4.2.6}$$

Thus we have the desired expression for the Fourier coefficients in terms of the signal $x(n)$.

The relationships (4.2.1) and (4.2.6) for the frequency analysis of discrete-time signals are summarized below.

FREQUENCY ANALYSIS OF DISCRETE-TIME PERIODIC SIGNALS

Synthesis equation	$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}$	(4.2.7)
Analysis equation	$c_k = \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N}$	(4.2.8)

Equation **(4.2.7)** is often called the ***discrete-rime Fourier series (DTFS).*** The Fourier coefficients $\{c_k\}$, $k = 0, 1, \ldots, N-1$ provide the description of $x(n)$ in the frequency domain. in the sense that c_k represents the amplitude and phase associated with the frequency component

$$s_k(n) = e^{j2\pi kn/N} = e^{j\omega_k n}$$

where $\omega_k = 2\pi k/N$.

We recall from Section **1.3.3** that the functions $s_k(n)$ are periodic with period **N.** Hence $s_k(n) = s_k(n + N)$. In view of this periodicity, it follows that the Fourier coefficients c_k, when viewed beyond the range $k = 0, 1, \ldots, N-1$, also satisfy a periodicity condition. Indeed, from (4.2.8), which holds for every value of k. we have

$$c_{k+N} = \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi(k+N)n/N} = \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} = c_k \tag{4.2.9}$$

Therefore, the Fourier series coefficients $\{c_k\}$ form a periodic sequence when extended outside of the range $k = 0, 1, \ldots, N-1$. Hence

$$c_{k+N} = c_k$$

that is, $\{c_k\}$ is a periodic sequence with fundamental period N. Thus ***the spectrum of a signal $x(n)$, which is periodic with period N, is a periodic sequence with period N.*** Consequently, any N consecutive samples of the signal or its spectrum provide a complete description of the signal in the time or frequency domains.

Although the Fourier coefficients form a periodic sequence, we will focus our attention on the single period with range $k = 0, 1, \ldots, N-1$. This is convenient, since in the frequency domain, this amounts to covering the fundamental range $0 \le \omega_k = 2\pi k/N < 2\pi$, for $0 \le k \le N-1$. *In* contrast, the frequency range $-\pi < \omega_k = 2\pi k/N \le \pi$, corresponds to $-N/2 < k \le N/2$, which creates an inconvenience when N is odd. Clearly. if we use a sampling frequency F_s, the range $0 \le k \le N-1$ corresponds to the frequency range $0 \le F < F_s$.

Example 4.2.1

Determine the spectra of the signals

(a) $x(n) = \cos\sqrt{2}\pi n$

(b) $x(n) = \cos \pi n/3$

(c) $x(n)$ is periodic with period $N = 4$ and

$$x(n) = \{\underset{\uparrow}{1}, 1, 0, 0\}$$

Solution

(a) For $\omega_0 = \sqrt{2}\pi$, **we have** $f_0 = 1/\sqrt{2}$. Since f_0 is not a rational number. the signal is not periodic. Consequently. this signal cannot be expanded in a Fourier series. Nevertheless. the signal does possess a spectrum. Its spectral content consists of the single frequency component at $\omega = \omega_0 = \sqrt{2}\pi$.

(b) In this case $f_0 = \frac{1}{6}$ and hence $x(n)$ is periodic with fundamental period $N = 6$. From (4.2.8) we have

$$c_k = \frac{1}{6}\sum_{n=0} x(n)e^{-j2\pi kn/6} \qquad k = 0, 1, \ldots, 5$$

However. $x(n)$ can be expressed as

$$x(n) = \cos\frac{2\pi n}{6} = \frac{1}{2}e^{j2\pi n/6} + \frac{1}{2}e^{-j2\pi n/6}$$

which is already in the form of the exponential Fourier series in **(4.2.7).** In comparing the two exponential terms in $x(n)$ with (4.2.7), it is apparent that $c_1 = \frac{1}{2}$. The second exponential in $x(n)$ corresponds to the term $k = -1$ in (4.2.7). However. this term can also be written as

$$e^{-j2\pi n/6} = e^{j2\pi(5-6)n/6} = e^{j2\pi(5n)/6}$$

which means that $c_{-1} = c_5$. But this is consistent with (4.2.9), and our previous observation that the Fourier series coefficients form a periodic sequence of period N. Consequently. we conclude that

$$c_0 = c_2 = c_3 = c_4 = 0$$
$$c_1 = \tfrac{1}{2} \qquad c_5 = \tfrac{1}{2}$$

(c) From (4.2.8). we have

$$c_k = \frac{1}{4}\sum_{n=0}^{3} x(n)e^{-j2\pi kn/4} \qquad k = 0, 1, 2, 3$$

or

$$c_k = \tfrac{1}{4}(1 + e^{-j\pi k/2}) \qquad k = 0, 1, 2, 3$$

For $k = 0, 1, 2, 3$ we obtain

$$c_0 = \tfrac{1}{2} \qquad c_1 = \tfrac{1}{4}(1 - j) \qquad c_2 = 0 \qquad c_3 = \tfrac{1}{4}(1 + j)$$

The magnitude and phase spectra are

$$|c_0| = \frac{1}{2} \qquad |c_1| = \frac{\sqrt{2}}{4} \qquad |c_2| = 0 \qquad |c_3| = \frac{\sqrt{2}}{4}$$

$$\measuredangle c_0 = 0 \qquad \measuredangle c_1 = -\frac{\pi}{4} \qquad \measuredangle c_2 = \text{undefined} \qquad \measuredangle c_3 = \frac{\pi}{4}$$

Figure 4.10 illustrates the spectral content of the signals in (b) and (c).

4.2.2 Power Density Spectrum of Periodic Signals

The average power of a discrete-time periodic signal with period N was defined in (2.1.23) as

$$P_x = \frac{1}{N}\sum_{n=0}^{N-1} |x(n)|^2 \tag{4.2.10}$$

We shall now derive an expression for P_x in terms of the Fourier coefficient $\{c_k\}$.

If we use the relation (4.2.7) in (4.2.10), we have

$$\begin{aligned} P_x &= \frac{1}{N}\sum_{n=0}^{N-1} x(n)x^*(n) \\ &= \frac{1}{N}\sum_{n=0}^{N-1} x(n)\left(\sum_{k=0}^{N-1} c_k^* e^{-j2\pi kn/N}\right) \end{aligned}$$

Now. we can interchange the order of the two summations and make use of (4.2.8). obtaining

$$\begin{aligned} P_x &= \sum_{k=0}^{N-1} c_k^* \left[\frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N}\right] \\ &= \sum_{k=0}^{N-1} |c_k|^2 = \frac{1}{N}\sum_{n=0}^{N-1} |x(n)|^2 \end{aligned} \tag{4.2.11}$$

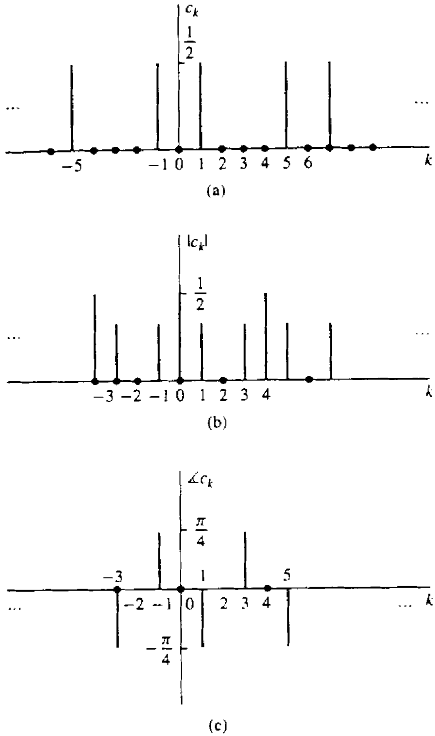


Figure 4.10 **Spectra of the periodic signals discussed** in **Example 4.2.1(b) and (c).**

which is the desired expression for the average power in the periodic signal. In other words, the average power in the signal is the sum of the powers of the individual frequency components. We view (4.2.11) as a Parseval's relation for discrete-time periodic signals. The sequence $|c_k|^2$ for $k = 0, 1, \ldots, N-1$ is the distribution of power as a function of frequency and is called the power **density** spectrum of the periodic signal.

If we are interested in the energy of the sequence $x(n)$ over a single period, (4.2.11) implies that

$$E_N = \sum_{n=0}^{N-1} |x(n)|^2 = N \sum_{k=0}^{N-1} |c_k|^2 \tag{4.2.12}$$

which is consistent with our previous results for continuous-time periodic signals.

If the signal $x(n)$ is real [i.e., $x^*(n) = x(n)$], then, proceeding as in Section 4.2.1, we can easily show that

$$c_k^* = c_{-k} \tag{4.2.13}$$

or equivalently,

$$|c_{-k}| = |c_k| \qquad \text{(even symmetry)} \tag{4.2.14}$$

$$-\measuredangle c_{-k} = \measuredangle c_k \qquad \text{(odd symmetry)} \tag{4.2.15}$$

These symmetry properties for the magnitude and phase spectra of a periodic signal, in conjunction with the periodicity property, have very important implications on the frequency range of discrete-time signals.

Indeed, by combining (4.2.9) with (4.2.14) and (4.2.15), we obtain

$$|c_k| = |c_{N-k}| \tag{4.2.16}$$

and

$$\measuredangle c_k = -\measuredangle c_{N-k} \tag{4.2.17}$$

More specifically, we have

$$\begin{aligned}
|c_0| &= |c_N|, & \measuredangle c_0 &= -\measuredangle c_N = 0 & & \\
|c_1| &= |c_{N-1}|, & \measuredangle c_1 &= -\measuredangle c_{N-1} & & \\
|c_{N/2}| &= |c_{N/2}|, & \measuredangle c_{N/2} &= 0 & & \text{if } N \text{ is even} \\
|c_{(N-1)/2}| &= |c_{(N+1)/2}|, & \measuredangle c_{(N-1)/2} &= -\measuredangle c_{(N+1)/2} & & \text{if } N \text{ is odd}
\end{aligned} \tag{4.2.18}$$

Thus, for a real signal, the spectrum c_k, $k = 0, 1, \ldots, N/2$ for N even, or $k = 0, 1, \ldots, (N-1)/2$ for N odd, completely specifies the signal in the frequency domain. Clearly, this is consistent with the fact that the highest relative frequency that can be represented by a discrete-time signal is equal to π. Indeed, if $0 \le \omega_k = 2\pi k/N \le \pi$, then $0 \le k \le N/2$.

By making use of these symmetry properties of the Fourier series coefficients of a real signal, the Fourier series in (4.2.7) can also be expressed in the alternative forms

$$x(n) = c_0 + 2\sum_{k=1}^{L} |c_k| \cos\left(\frac{2\pi}{N}kn + \theta_k\right) \tag{4.2.19}$$

$$= a_0 + \sum_{k=1}^{L}\left(a_k \cos\frac{2\pi}{N}kn - b_k \sin\frac{2\pi}{N}kn\right) \tag{4.2.20}$$

where $a_0 = c_0$, $a_k = 2|c_k|\cos\theta_k$, $b_k = 2|c_k|\sin\theta_k$, and $L = N/2$ if N is even and $L = (N-1)/2$ if N is odd.

Finally, we note that as in the case of continuous-time signals, the power density spectrum $|c_k|^2$ does not contain any phase information. Furthermore, the spectrum is discrete and periodic with a fundamental period equal to that of the signal itself.

Example 4.2.2 Periodic "Square-Wave" Signal

Determine the Fourier series coefficients and the power density spectrum of the periodic signal shown in Fig. 4.11.

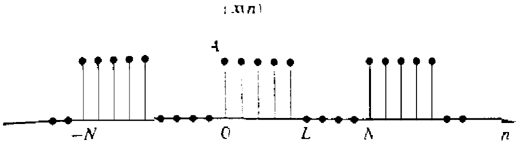


Figure 4.11 Discrete-time periodic square-wave signal

Solution By applying the analysis equation (4.2.8) to the signal shown in Fig. 4.11. we obtain

$$c_k = \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} = \frac{1}{N}\sum_{n=0}^{L-1} Ae^{-j2\pi kn/N} \qquad k = 0, 1, \ldots, N-1$$

which is a geometric summation. Now we can use (4.2.3) to simplify the summation above. Thus we obtain

$$c_k = \frac{A}{N}\sum_{n=0}^{L-1}(e^{-j2\pi k/N})^n = \begin{cases} \dfrac{AL}{N}, & k = 0 \\ \dfrac{A}{N}\dfrac{1 - e^{-j2\pi kL/N}}{1 - e^{-j2\pi k/N}}, & k = 1, 2, \ldots, N-1 \end{cases}$$

The last expression can be simplified further if we note that

$$\frac{1 - e^{-j2\pi kL/N}}{1 - e^{-j2\pi k/N}} = \frac{e^{-j\pi kL/N}}{e^{-j\pi k/N}}\,\frac{e^{j\pi kL/N} - e^{-j\pi kL/N}}{e^{j\pi k/N} - e^{-j\pi k/N}}$$

$$= e^{-j\pi k(L-1)/N}\frac{\sin(\pi kL/N)}{\sin(\pi k/N)}$$

Therefore.

$$c_k = \begin{cases} \dfrac{AL}{N}, & k = 0, +N, \pm 2N, \\ \dfrac{A}{N}e^{-j\pi k(L-1)/N}\dfrac{\sin(\pi kL/N)}{\sin(\pi k/N)}, & \text{otherwise} \end{cases} \tag{4.2.21}$$

The power density spectrum of this periodic signal is

$$|c_k|^2 = \begin{cases} \left(\dfrac{AL}{N}\right)^2, & k = 0, +N, \pm 2N, \ldots \\ \left(\dfrac{A}{N}\right)^2\left(\dfrac{\sin \pi kL/N}{\sin \pi k/N}\right)^2, & \text{otherwise} \end{cases} \tag{4.2.22}$$

Figure 4.12 illustrates the plots of $|c_k|^2$ for L = 5 and 7, N = 40 and 60. and A = 1

4.2.3 The Fourier Transform of Discrete-Time Aperiodic *Signals*

Just as in the case of continuous-time aperiodic energy signals, the frequency analysis of discrete-time aperiodic finite-energy signals involves a Fourier transform of the time-domain signal. Consequently, the development in this section parallels to a large extent, that given in Section 4.1.3.

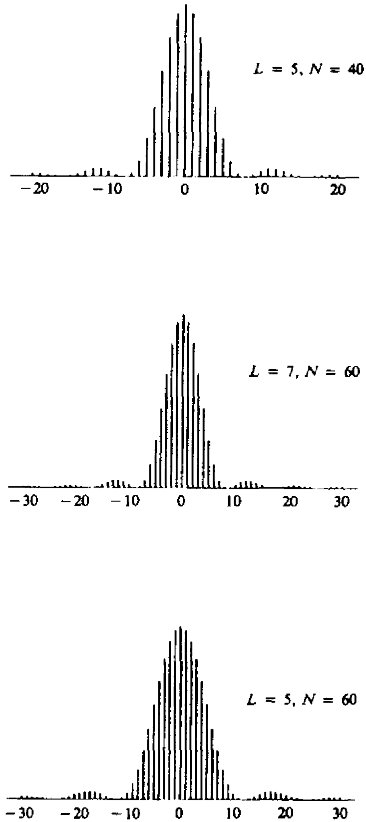


Figure 4.12 Plot of the power density spectrum given by (4.2.22).

The Fourier transform of a finite-energy discrete-time signal $x(n)$ is defined as

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \tag{4.2.23}$$

Physically. $X(\omega)$ represents the frequency content of the signal $x(n)$. In other words, $X(\omega)$ is a decomposition of $x(n)$ into its frequency components.

We observe two basic differences between the Fourier transform of a discrete-time finite-energy signal and the Fourier transform of a finite-energy analog signal. First. for continuous-time signals, the Fourier transform, and hence the spectrum of the signal, have a frequency range of $(-\infty, \infty)$. In contrast, the frequency range for a discrete-time signal is unique over the frequency interval of $(-\pi, \pi)$ or. equivalently, $(0, 2\pi)$. This property is reflected in the Fourier transform of the signal. Indeed. $X(\omega)$ is periodic with period 2π, that is.

$$\begin{aligned} X(\omega+2\pi k) &= \sum_{n=-\infty}^{\infty} x(n)e^{-j(\omega+2\pi k)n} \\ &= \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}e^{-j2\pi kn} \\ &= \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = X(\omega) \end{aligned} \tag{4.2.24}$$

Hence $X(w)$ is periodic with period 2π. But this property is just a consequence of the fact that the frequency range for any discrete-time signal is limited to $(-\pi, \pi)$ or $(0, 2\pi)$, and any frequency outside this interval is equivalent to a frequency within the interval.

The second basic difference is also a consequence of the discrete-time nature of the signal. Since the signal is discrete in time. the Fourier transform of the signal involves a summation of terms instead of an integral, as in the case of continuous-time signals.

Since $X(\omega)$ is a periodic function of the frequency variable ω, it has a Fourier series expansion, provided that the conditions for the existence of the Fourier series, described previously, are satisfied. In fact, from the definition of the Fourier transform $X(\omega)$ of the sequence $x(n)$, given by (4.2.23), we observe that $X(\omega)$ has the form of a Fourier series. The Fourier coefficients in this series expansion are the values of the sequence $x(n)$.

To demonstrate this point. let us evaluate the sequence $x(n)$ from $X(\omega)$. First, we multiply both sides **(4.2.23)** by $e^{j\omega m}$ and integrate over the interval $(-\pi, \pi)$. Thus we have

$$\int_{-\pi}^{\pi} X(\omega)e^{j\omega m}d\omega = \int_{-\pi}^{\pi}\left[\sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}\right]e^{j\omega m}d\omega \tag{4.2.25}$$

The integral on the right-hand side of (4.2.25) can **be** evaluated if we can interchange the order of summation and integration. This interchange can be made if the series

$$X_N(\omega) = \sum_{n=-N}^{N} x(n)e^{-j\omega n}$$

converges uniformly to $X(\omega)$ as $N \to \infty$. Uniform convergence means that, for every ω, $X_N(\omega) \to X(\omega)$, as $N \to \infty$. The convergence of the Fourier transform is discussed in more detail in the following section. For the moment, let us assume that the series converges uniformly, so that we can interchange the order of summation and integration in (4.2.25). Then

$$\int_{-\pi}^{\pi} e^{j\omega(m-n)}d\omega = \begin{cases} 2\pi, & m = n \\ 0, & m \neq n \end{cases}$$

Consequently,

$$\sum_{n=-\infty}^{\infty} x(n) \int_{-\pi}^{\pi} e^{j\omega(m-n)} d\omega = \begin{cases} 2\pi x(m), & m = n \\ 0, & m \neq n \end{cases} \tag{4.2.26}$$

By combining **(4.2.25)** and **(4.2.26)**, **we** obtain the desired result that

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega \tag{4.2.27}$$

If we compare the integral in **(4.2.27)** with (4.1.9), we note that this is just the expression for the Fourier series coefficient for a function that is periodic with period 2π. The only difference between **(4.1.9)** and **(4.2.27)** is the sign on the exponent in the integrand, which is a consequence of our definition of the Fourier transform as given by (4.2.23). Therefore, the Fourier transform of the sequence $x(n)$, defined by (4.2.23), has the form of a Fourier series expansion.

In summary, the *Fourier transform pair for discrete-time signals* is as follows.

FREQUENCY ANALYSIS OF DISCRETE-TIME APERIODIC SIGNALS

Synthesis **equation** **inverse transform**	$x(n) = \frac{1}{2\pi} \int_{2\pi} X(\omega) e^{j\omega n} d\omega$	(4.2.28)
Analysis equation **direct transform**	$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$	(4.2.29)

4.2.4 Convergence of the Fourier Transform

In the derivation of the inverse transform given by (4.2.28), **we** assumed that the series

$$X_N(\omega) = \sum_{n=-N}^{N} x(n) e^{-j\omega n} \tag{4.2.30}$$

converges uniformly to $X(\omega)$, given in the integral of (4.2.28), as $N \to \infty$. By uniform convergence we mean that for each ω,

$$\lim_{N\to\infty} \left\{\sup_{\omega} |X(\omega) - X_N(\omega)|\right\} = 0 \tag{4.2.31}$$

Uniform convergence is guaranteed if $x(n)$ is absolutely summable. Indeed, if

$$\sum_{n=-\infty}^{\infty} |x(n)| < \infty \tag{4.2.32}$$

then

$$|X(\omega)| = \left| \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \right| \leq \sum_{n=-\infty}^{\infty} |x(n)| < \infty$$

Hence (4.2.32) is a sufficient condition for the existence of the discrete-time Fourier transform. We note that this is the discrete-time counterpart of the third Dirichlet condition for the Fourier transform of continuous-time signals. The first two conditions do not apply due to the discrete-time nature of $\{x(n)\}$.

Some sequences are not absolutely summable, but they are square summable. That is, they have finite energy

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 < \infty \tag{4.2.33}$$

which is a weaker condition than (4.2.32). We would like to define the Fourier transform of finite-energy sequences, but we must relax the condition of uniform convergence. For such sequences we can impose a mean-square convergence condition:

$$\lim_{N\to\infty} \int_{-\pi}^{\pi} |X(\omega) - X_N(\omega)|^2 d\omega = 0 \tag{4.2.34}$$

Thus the energy in the error $X(\omega) - X_N(\omega)$ tends toward zero, but the error $|X(\omega) - X_N(\omega)|$ does not necessarily tend to zero. In this way we can include finite-energy signals in the class of signals for which the Fourier transform exists.

Let us consider an example from the class of finite-energy signals. Suppose that

$$X(\omega) = \begin{cases} 1, & |\omega| \le \omega_c \\ 0, & \omega_c < |\omega| \le \pi \end{cases} \tag{4.2.35}$$

The reader should remember that $X(\omega)$ is periodic with period 2π. Hence **(4.2.35)** represents only one period of $X(\omega)$. The inverse transform of $X(\omega)$ results in the sequence

$$\begin{aligned} x(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \frac{\sin \omega_c n}{\pi n} \qquad n \ne 0 \end{aligned}$$

For $n = 0$, we have

$$x(0) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} d\omega = \frac{\omega_c}{\pi}$$

Hence

$$x(n) = \begin{cases} \dfrac{\omega_c}{\pi}, & n = 0 \\ \dfrac{\omega_c}{\pi} \dfrac{\sin \omega_c n}{\omega_c n}, & n \ne 0 \end{cases} \tag{4.2.36}$$

This transform pair is illustrated in Fig. **4.13.**

Sometimes, the sequence $\{x(n)\}$ in **(4.2.36)** is expressed as

$$x(n) = \frac{\sin \omega_c n}{\pi n} \qquad -\infty < n < \infty \tag{4.2.37}$$

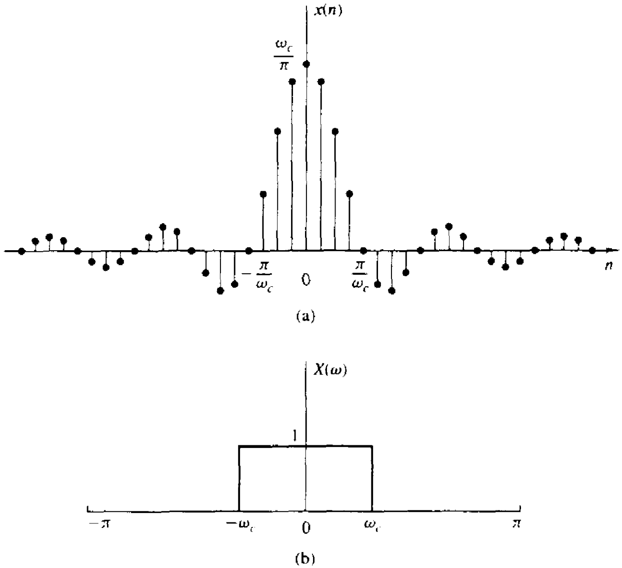


Figure 4.13 Fourier transform pair in (4.2.35) and (4.2.36).

with the understanding that at $n = 0$, $x(n) = \omega_c/\pi$. We should emphasize, however, that $(\sin \omega_c n)/\pi n$ is not a continuous function, and hence L'Hospital's rule cannot be used to determine $x(0)$.

Now let us consider the determination of the Fourier transform of the sequence given by (4.2.37). The sequence $\{x(n)\}$ is not absolutely summable. Hence the infinite series

$$\sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \frac{\sin \omega_c n}{\pi n} e^{-j\omega n} \tag{4.2.38}$$

does not converge uniformly for all ω. However, the sequence $\{x(n)\}$ has a finite energy $E_x = \omega_c/\pi$ as will be shown in Section **4.3.** Hence the sum in (4.2.38) is guaranteed to converge to the $X(\omega)$ given by (4.2.35) in the mean-square sense.

To elaborate on this point, let us consider the finite sum

$$X_N(\omega) = \sum_{n=-N}^{N} \frac{\sin \omega_c n}{\pi n} e^{-j\omega n} \tag{4.2.39}$$

Figure **4.14** shows the function $X_N(\omega)$ for several values of N. We note that there is a significant oscillatory overshoot at $\omega = \omega_c$, independent of the value of N. As

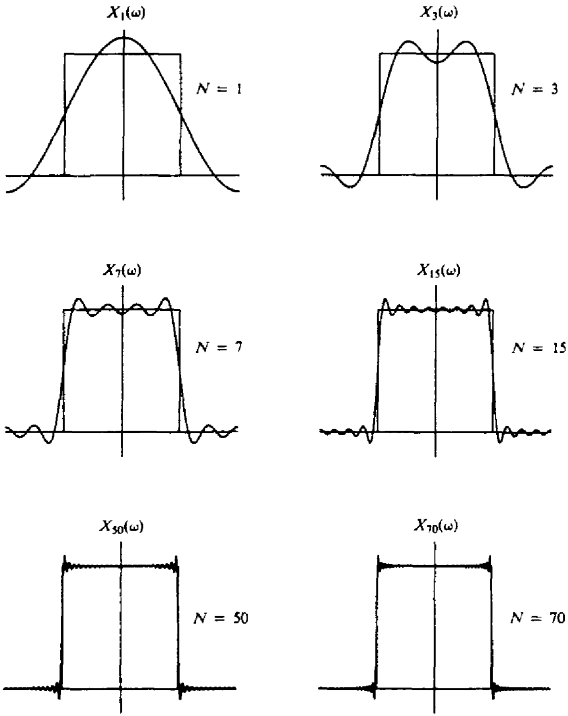


Figure 4.14 Illustration of convergence of the Fourier transform and the Gibbs phenomenon at the point of discontinuity.

N increases, the oscillations become more rapid. but the size of the ripple remains the same. One can show that as $N \to \infty$, the oscillations converge to the point of the discontinuity at $\omega = \omega_c$, but their amplitude does not go to zero. However, (4.2.34) is satisfied, and therefore $X_N(\omega)$ converges to $X(\omega)$ in the mean-square sense.

The oscillatory behavior of the approximation $X_N(\omega)$ to the function $X(\omega)$ at a point of discontinuity of $X(\omega)$ is called the **Gibbs** phenomenon. A similar effect is observed in the truncation of the Fourier series of a continuous-time periodic signal, given by the synthesis equation (4.1.8). For example, the truncation of the Fourier series for the periodic square-wave signal in Example 4.1.1, gives rise to the same oscillatory behavior in the finite-sum approximation of $x(t)$. The Gibbs phenomenon will be encountered again in the design of practical, discrete-time FIR systems considered in Chapter 8.

4.2.5 Energy Density Spectrum of Aperiodic Signals

Recall that the energy of a discrete-time signal $x(n)$ is defined as

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 \tag{4.2.40}$$

Let us now express the energy E_x in terms of the spectral characteristic $X(\omega)$. First we have

$$E_x = \sum_{n=-\infty}^{\infty} x(n)x^*(n) = \sum_{n=-\infty}^{\infty} x(n)\left[\frac{1}{2\pi}\int_{-\pi}^{\pi} X^*(\omega)e^{-j\omega n}\,d\omega\right]$$

If we interchange the order of integration and summation in the equation above, we obtain

$$\begin{aligned} E_x &= \frac{1}{2\pi}\int_{-\pi}^{\pi} X^*(\omega)\left[\sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}\right]d\omega \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} |X(\omega)|^2 d\omega \end{aligned}$$

Therefore. the energy relation between $x(n)$ and $X(\omega)$ is

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |X(\omega)|^2 d\omega \tag{4.2.41}$$

This is Parseval's relation for discrete-time aperiodic signals with finite energy.

The spectrum $X(\omega)$ is. in general, a complex-valued function of frequency. It may be expressed as

$$X(\omega) = |X(\omega)|e^{j\Theta(\omega)} \tag{4.2.42}$$

where

$$\Theta(\omega) = \measuredangle X(\omega)$$

is the phase spectrum and $|X(\omega)|$ is the magnitude spectrum.

As in the case of continuous-time signals, the quantity

$$S_{xx}(\omega) = |X(\omega)|^2 \tag{4.2.43}$$

represents the distribution of energy as a function of frequency, and it is called the energy *density spectrum* of $x(n)$. Clearly, $S_{xx}(\omega)$ does not contain any phase information.

Suppose now that the signal $x(n)$ is real. Then it easily follows that

$$X^*(\omega) = X(-\omega) \tag{4.2.44}$$

or equivalently,

$$|X(-\omega)| = |X(\omega)| \qquad \text{(even symmetry)} \tag{4.2.45}$$

and

$$\measuredangle X(-\omega) = -\measuredangle X(\omega) \qquad \text{(odd symmetry)} \tag{4.2.46}$$

From (4.2.43) it also follows that

$$S_{xx}(-\omega) = S_{xx}(\omega) \qquad \textbf{(even} \text{ symmetry)} \tag{4.2.47}$$

From these symmetry properties we conclude that the frequency range of real discrete-time signals can be limited further to the range $0 \leq \omega \leq \pi$ (i.e., one-half of the period). Indeed, if we know $X(\omega)$ in the range $0 \leq \omega \leq \pi$, we can determine it for the range $-\pi \leq \omega < 0$ using the symmetry properties given above. As we have already observed. similar results hold for discrete-time periodic signals. Therefore. the frequency-domain description of a real discrete-time signal is completely specified by its spectrum in the frequency range $0 \leq$ w $\leq \pi$.

Usually, we work with the fundamental interval $0 \leq$ w $\leq \pi$ or $0 \leq$ F $\leq F_s/2$, expressed in Hertz. We sketch more than half a period only when required by the specific application.

Example 4.2.3

Derermine and sketch the energy densiry spectrum $S_{xx}(\omega)$ of the signal

$$x(n) = \text{i} \quad \text{t} \quad) \qquad -1 < a < 1$$

Solution Since $|a| < 1$. the sequence $x(n)$ is absolutely summahle, as can he verified by applying the geometric summation formula.

$$\sum_{n=-\infty}^{\infty} |x(n)| = \sum_{n=0}^{\infty} |a|^n = \frac{1}{1-|a|} < \infty$$

Hence the Fourier transform of $x(n)$ exists and is obtained by applying (4.2.29). Thus

$$X(\omega) = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n$$

Since $|ae^{-j\omega}| = |a| < 1$. use of the geometric summation formula again yields

$$X(\omega) = \frac{1}{1-ae^{-j\omega}}$$

The energy density spectrum is given by

$$S_{xx}(\omega) = |X(\omega)|^2 = X(\omega)X^*(\omega) = \frac{1}{(1-ae^{-j\omega})(1-ae^{j\omega})}$$

or. equivalently, as

$$S_{xx}(\omega) = \frac{1}{1-2a\cos\omega + a^2}$$

Note that $S_{xx}(-\omega) = S_{xx}(\omega)$ *in* accordance *with* **(4.2.47).**

Figure 4.15 shows the signal $x(n)$ and **its** corresponding spectrum for a = 0.5 **and** a = –0.5. Note that for ***a*** = –0.5 the signal has more rapid variations and as a result its spectrum has stronger high frequencies.

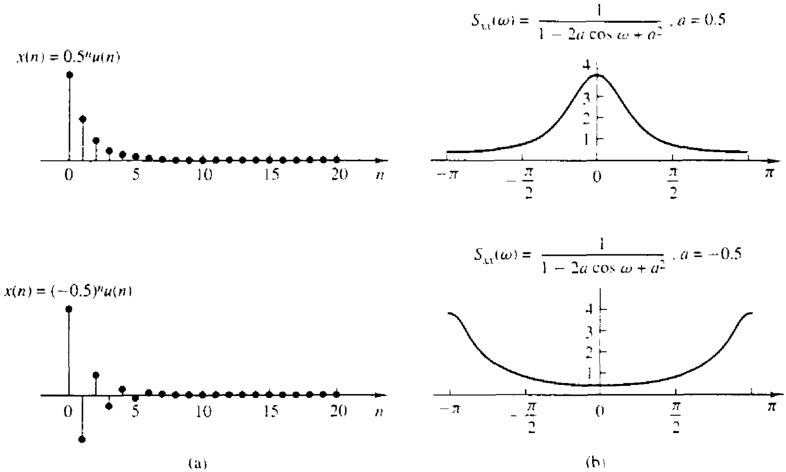


Figure 4.11 (a) Sequence $x(n) = (\frac{1}{2})^n u(n)$ and $x(n) = (-\frac{1}{2})^n u(n)$; (b) their energy density spectra.

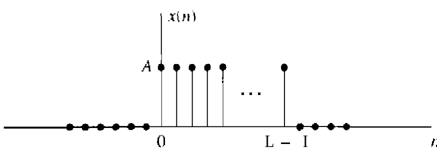


Figure 4.16 Discrete-time rectangular pulse.

Example 4.2.4

Determine the Fourier transform and the energy density spectrum of the sequence

$$x(n) = \begin{cases} A, & 0 \le n \le L-1 \\ 0, & \text{otherwise} \end{cases} \tag{4.2.48}$$

which is illustrated in Fig. 4.16.

Solution Before computing the Fourier transform, we observe that

$$\sum_{n=-\infty}^{\infty} |x(n)| = \sum_{n=0}^{L-1} |A| = L|A| < \infty$$

Hence $x(n)$ is absolutely summable and its Fourier transform exists. Furthermore, we note that $x(n)$ is a finite-energy signal with $E_x = |A|^2 L$.

The Fourier transform of this signal is

$$X(\omega) = \sum^{L-1} Ae^{-j\omega n}$$

$$= A\frac{1-e^{-j\omega L}}{1-e^{-j\omega}}$$

$$= Ae^{-j(\omega/2)(L-1)}\frac{\sin(\omega L/2)}{\sin(\omega/2)} \tag{4.2.49}$$

For $\omega = 0$ the transform in (4.2.49) yields $X(0) = AL$, which is easily established by setting $\omega = 0$ in the defining equation for $X(\omega)$, or by using L'Hospital's rule in (4.2.49) to resolve the indeterminate form when $\omega = 0$.

The magnitude and phase spectra of $x(n)$ are

$$|X(\omega)| = \begin{cases} |A|L, & \omega = 0 \\ |A|\left|\dfrac{\sin(\omega L/2)}{\sin(\omega/2)}\right|, & \text{otherwise} \end{cases} \tag{4.2.50}$$

and

$$\measuredangle X(\omega) = \measuredangle A - \frac{\omega}{2}(L-1) + \measuredangle \frac{\sin(\omega L/2)}{\sin(\omega/2)} \tag{4.2.51}$$

where we should remember that the phase of a real quantity is zero if the quantity is positive and π if it is negative.

The spectra $|X(\omega)|$ and $\measuredangle X(\omega)$ are shown in Fig. 4.17 for the case $A = 1$ and $L = 5$. The energy density spectrum is simply the square of the expression given in (4.2.50).

There is an interesting relationship that exists between the Fourier transform of the constant amplitude pulse in Example 4.2.4 and the periodic rectangular

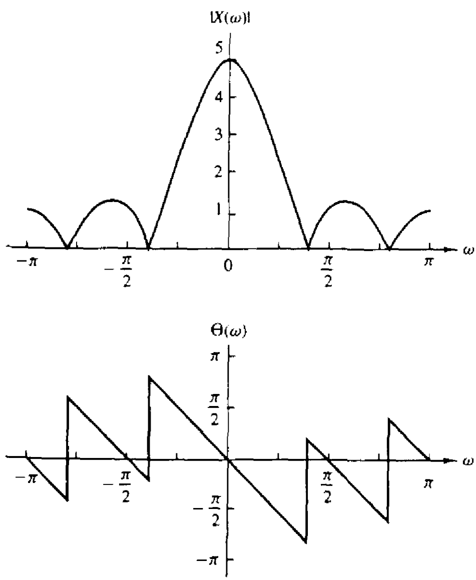


Figure 4.17 Magnitude and phase of Fourier transform of the discrete-time rectangular pulse in Fig. 4.16.

wave considered in Example 4.2.2. If we evaluate the Fourier transform as given in (4.2.49) at a set of equally spaced (harmonically related) frequencies

$$\omega_k = \frac{2\pi}{N}k \qquad k = 0, 1, \ldots, N-1$$

we obtain

$$X\left(\frac{2\pi}{N}k\right) = Ae^{-j(\pi/N)k(L-1)}\frac{\sin[(\pi/N)kL]}{\sin[(\pi/N)k]} \tag{4.2.52}$$

If we compare this result with the expression for the Fourier series coefficients given in (4.2.21) for the periodic rectangular wave, we find that

$$X\left(\frac{2\pi}{N}k\right) = Nc_k \qquad k = 0, 1, \ldots, N-1 \tag{4.2.53}$$

To elaborate, we have established that the Fourier transform of the rectangular pulse, which is identical with a single period of the periodic rectangular pulse train, evaluated at the frequencies $\omega = 2\pi k/N$, $k = 0, 1, \ldots, N-1$, which are identical to the harmonically related frequency components used in the Fourier series representation of the periodic signal, is simply a multiple of the Fourier coefficients $\{c_k\}$ at the corresponding frequencies.

The relationship given in (4.2.53) for the Fourier transform of the rectangular pulse evaluated at $\omega = 2\pi k/N$, $k = 0, 1, \ldots, N-1$, and the Fourier coefficients of the corresponding periodic signal, is not only true for these two signals but, in fact, holds in general. This relationship is developed further in Chapter 5.

4.2.6 Relationship of the Fourier Transform to the z-Transform

The z-transform of a sequence $x(n)$ is defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} \qquad \textbf{ROC: } r_2 < |z| < r_1 \tag{4.2.54}$$

where $r_2 < |z| < r_1$ is the region of convergence of $X(z)$. Let us express the complex variable z in polar form as

$$z = re^{j\omega} \tag{4.2.55}$$

where $r = |z|$ and $\omega = \measuredangle z$. Then, within the region of convergence of $X(z)$, we can substitute $z = re^{j\omega}$ into (4.2.54). This yields

$$X(z)|_{z=re^{j\omega}} = \sum_{n=-\infty}^{\infty} [x(n)r^{-n}]e^{-j\omega n} \tag{4.2.56}$$

From the relationship in (4.2.56) we note that $X(z)$ can be interpreted as the Fourier transform of the signal sequence $x(n)r^{-n}$. The weighting factor r^{-n} is growing with n if $r < 1$ and decaying if $r > 1$. Alternatively, if $X(z)$ converges for

$|z| = 1$, then

$$X(z)|_{z=e^{j\omega}} \equiv X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \tag{4.2.57}$$

Therefore, the Fourier transform can be viewed as the z-transform of the sequence evaluated on the unit circle. If $X(z)$ does not converge in the region $|z| = 1$ [i.e.. if the unit circle is not contained in the region of convergence of $X(z)$], the Fourier transform $X(\omega)$ does not exist.

We should note that the existence of the z-transform requires that the sequence $\{x(n)r^{-n}\}$ be absolutely summable for some value of r. that is.

$$\sum_{n=-\infty}^{\infty} |x(n)r^{-n}| < \infty \tag{4.2.58}$$

Hence if (4.2.58) converges only for values of $r > r_0 > 1$. the z-transform exists. but the Fourier transform does not exist. This is the case. for example. for causal sequences of the form $x(n) = a^n u(n)$, where $|a| > 1$.

There are sequences. however. that do not satisfy the requirement in (4.2.58). for example, the sequence

$$x(n) = \frac{\sin \omega_c n}{\pi n} \qquad -\infty < n < \infty \tag{4.2.59}$$

This sequence does not have a z-transform. Since it has a finite energy. its Fourier transform converges in the mean-square sense to the discontinuous function $X(\omega)$. defined as

$$X(\omega) = \begin{cases} 1, & |\omega| < \omega_c \\ 0, & \omega_c < |\omega| \le \pi \end{cases} \tag{4.2.60}$$

In conclusion. the existence of the z-transform requires that (4.2.58) be satisfied for some region in the z-plane. If this region contains the unit circle. the Fourier transform $X(\omega)$ exists. However, the existence of the Fourier transform. which is defined for finite energy signals. does not necessarily ensure the existence of the z-transform.

4.2.7 The Cepstrum

Let us consider a sequence $\{x(n)\}$ having a z-transform $X(z)$. We assume that $\{x(n)\}$ is a stable sequence so that $X(z)$ converges on the unit circle. The *complex cepstrum* of the sequence $\{x(n)\}$ is defined as the sequence $\{c_x(n)\}$, which is the inverse z-transform of $C_x(z)$, where

$$C_x(z) = \ln X(z) \tag{4.2.61}$$

The complex cepstrum exists if $C_x(z)$ converges in the annular region $r_1 < |z| < r_2$, where $0 < r_1 < 1$ and $r_2 > 1$. Within this region of convergence. $C_x(z)$ can be represented by the Laurent series

$$C_x(z) = \ln X(z) = \sum_{n=-\infty}^{\infty} c_x(n)z^{-n} \tag{4.2.62}$$

where

$$c_x(n) = \frac{1}{2\pi j}\int_C \ln X(z)z^{n-1}dz \tag{4.2.63}$$

C is a closed contour about the origin and lies within the region of convergence, Clearly. if $C_x(z)$ can be represented as in (4.2.62), the complex cepstrum sequence $\{c_x(n)\}$ is stable. Furthermore, if the complex cepstrum exists, $C_x(z)$ converges on the unit circle and hence we have

$$C_x(\omega) = \ln X(\omega) = \sum_{n=-\infty}^{\infty} c_x(n)e^{-j\omega n} \tag{4.2.64}$$

where $\{c_x(n)\}$ is the sequence obtained from the inverse Fourier transform of $\ln X(\omega)$, that is,

$$c_x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \ln X(\omega)e^{j\omega n}d\omega \tag{4.2.65}$$

If we express $X(\omega)$ in terms of its magnitude and phase, say

$$X(\omega) = |X(\omega)|e^{j\theta(\omega)} \tag{4.2.66}$$

then

$$\ln X(\omega) = \ln|X(\omega)| + j\theta(\omega) \tag{4.2.67}$$

By substituting **(4.2.67)** into (4.2.65), we obtain the complex cepstrum in the form

$$c_x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} [\ln|X(\omega)| + j\theta(\omega)]e^{j\omega n}d\omega \tag{4.2.68}$$

We can separate the inverse Fourier transform in **(4.2.68)** into the inverse Fourier transforms of $\ln|X(\omega)|$ and $\theta(\omega)$:

$$c_m(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \ln|X(\omega)|e^{j\omega n}d\omega \tag{4.2.69}$$

$$c_\theta(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \theta(\omega)e^{j\omega n}d\omega \tag{4.2.70}$$

In some applications, such as speech signal processing, only the component $c_m(n)$ is computed. In such a case the phase of $X(\omega)$ is ignored. Therefore. the sequence $\{x(n)\}$ cannot be recovered from $\{c_m(n)\}$. That is, the transformation from $\{x(n)\}$ to $\{c_m(n)\}$ is not invertible.

In speech signal processing, the (real) cepstrum has been used to separate and thus to estimate the spectral content of the speech from the pitch frequency of the speech. The complex cepstrum is used in practice to separate signals that are convolved. The process of separating two convolved signals is called *deconvolution* and the use of the complex cepstrurn to perform the separation is called *homomorphic deconvolution*. This topic is discussed in Section **4.6.**

4.2.8 The Fourier Transform of Signals with Poles on the Unit Circle

As was shown in Section 4.2.6, the Fourier transform of a sequence $x(n)$ can be determined by evaluating its z-transform $X(z)$ on the unit circle, provided that the unit circle lies within the region of convergence of $X(z)$. Otherwise, the Fourier transform does not exist.

There are some aperiodic sequences that are neither absolutely summable nor square summable. Hence their Fourier transforms do not exist. One such sequence is the unit step sequence. which has the z-transform

$$X(z) = \frac{1}{1 - z^{-1}}$$

Another such sequence is the causal sinusoidal signal sequence $x(n) = (\cos \omega_0 n)u(n)$. This sequence has the z-transform

$$X(z) = \frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$$

Note that both of these sequences have poles on the unit circle.

For sequences such as these two examples, it is sometimes useful to extend the Fourier transform representation. This can be accomplished, in a mathematically rigorous way, by allowing the Fourier transform to contain impulses at certain frequencies corresponding to the location of the poles of $X(z)$ that lie on the unit circle. The impulses are functions of the continuous frequency variable ω and have infinite amplitude, zero width, and unit area. An impulse can be viewed as the limiting form of a rectangular pulse of height $1/a$ and width a, in the limit as $a \to 0$. Thus, by allowing impulses in the spectrum of a signal, it is possible to extend the Fourier transform representation to some signal sequences that are neither absolutely summable nor square summable.

The following example illustrates the extension of the Fourier transform representation for three sequences.

Example 4.2.5

Determine the Fourier transform of the following signals.

(a) $x_1(n) = u(n)$
(b) $x_2(n) = (-1)^n u(n)$
(c) $x_3(n) = (\cos \omega_0 n)u(n)$

by evaluating their z-transforms on the unit circle.

Solution

(a) From **Table** 4.3 we find that

$$X_1(z) = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1} \qquad \textbf{ROC: } |z| > 1$$

$X_1(z)$ has a pole, $p_1 = 1$, on the unit circle, but convenges for $|z| > 1$.

If we evaluate $X_1(z)$ on the unit circle. except at $z = 1$. we obtain

$$X_1(\omega) = \frac{e^{j\omega/2}}{2j\sin(\omega/2)} = \frac{1}{2\sin(\omega/2)} e^{j(\omega-\pi/2)} \qquad \omega \neq 2\pi k \quad k = 0, 1, \ldots$$

At $\omega = 0$ and multiples of 271. $X_1(\omega)$ contains impulses of area π.

Hence the presence of a pole at $z = 1$ (i.e., at $\omega = 0$) creates a problem only when we want to compute $|X_1(\omega)|$ at $\omega = 0$. because $|X_1(\omega)| \to \infty$ as $w \to 0$. For any other value of w. $X_1(\omega)$ is finite (i.e., well behaved). Although, at first glance one might expect the signal to have zero-frequent! components at all frequencies except at $\omega = 0$. this is not the case. This happens because the signal $x_1(n)$ is not a constant for all $-\infty < n < \infty$. Instead, it is *turned on* at $n = 0$. This abrupt jump creates all frequency components existing in the range $0 < \omega \leq \pi$. Generally. all signals which start at a finite time have nonzero-frequency components everywhere in the frequency axis from zero up to the folding frequency.

(b) From Table 3.3 we find that the z-transform of $a^n u(n)$ with $a = -1$ reduces to

$$X_2(z) = \frac{1}{1 + z^{-1}} = \frac{z}{z+1} \qquad \textbf{ROC: } |z| > 1$$

which has a pole at $z = -1 = e^{j\pi}$. The Fourier transform evaluated at frequencies other than $\omega = \pi$ and multiples of 2π is

$$X_2(\omega) = \frac{e^{j\omega/2}}{2\cos(\omega/2)} \qquad \omega \neq 2\pi(k + \tfrac{1}{2}) \quad k = 0, 1,$$

In this case the impulses occurs at $\omega = \pi + 2\pi k$.

Hence the magnitude is

$$|X_2(\omega)| = \frac{1}{2|\cos(\omega/2)|} \qquad \omega \neq 2\pi k + \pi \quad k = 0, 1 \ldots$$

and the phase is

$$\measuredangle X_2(\omega) = \begin{cases} \dfrac{\omega}{2}, & \text{if } \cos\dfrac{\omega}{2} \geq 0 \\ \dfrac{\omega}{2} + \pi, & \text{if } \cos\dfrac{\omega}{2} < 0 \end{cases}$$

Note that due to the presence of the pole at a $= -1$ (i.e., at frequency w $= \pi$), the magnitude of the Fourier transform becomes infinite. Now $|X(\omega)| \to \infty$ as $\omega \to \pi$. We observe that $(-1)^n u(n) = (\cos \pi n) u(n)$, which is the fastest possible oscillating signal in discrete time.

(c) From the discussion above. it follows that $X_3(\omega)$ is infinite at the frequency component w $= \omega_0$. Indeed, from Table 3.3. we find that

$$x_3(n) = (\cos\omega_0 n) u(n) \longleftrightarrow X_3(z) = \frac{1 - z^{-1}\cos\omega_0}{1 - 2z^{-1}\cos\omega_0 + z^{-2}} \qquad \textbf{ROC: } |z| > 1$$

The Fourier transform is

$$X_3(\omega) = \frac{1 - e^{-j\omega}\cos\omega_0}{(1 - e^{-j(\omega-\omega_0)})(1 - e^{j(\omega+\omega_0)})} \qquad \omega \neq \pm\omega_0 + 2\pi k \quad k = 0, 1, \ldots$$

The magnitude of $X_3(\omega)$ **is given by**

$$|X_3(\omega)| = \frac{|1 - e^{-j\omega}\cos\omega_0|}{|1 - e^{-j(\omega-\omega_0)}||1 - e^{-j(\omega+\omega_0)}|} \qquad \omega \neq \pm\omega_0 + 2\pi k \quad k = 0, 1.$$

Now if $\omega = -\omega_0$ or $\omega = \omega_0$, $|X_3(\omega)|$ becomes **infinite. For all other frequencies. the Fourier transform is well behaved.**

4.2.9 The Sampling Theorem Revisited

To process a continuous-time signal using digital signal processing techniques, it is necessary to convert the signal into a sequence of numbers. As was discussed in Section 1.4, this is usually done by sampling the analog signal, say $x_a(t)$, periodically every T seconds to produce a discrete-time signal $x(n)$ given by

$$x(n) = x_a(nT) \qquad -\infty < n < \infty \tag{4.2.71}$$

The relationship (4.2.71) describes the sampling process in the time domain. As discussed in Chapter 1, the sampling frequency $F_s = 1/T$ must be selected large enough such that the sampling does not cause any loss of spectral information (no aliasing). Indeed. if the spectrum of the analog signal can be recovered from the spectrum of the discrete- time signal, there is no loss of information. Consequently, we investigate the sampling process by finding the relationship between the spectra of signals $x_a(t)$ and $x(n)$.

If $x_a(t)$ is an aperiodic signal with finite energy. its (voltage) spectrum is given by the Fourier transform relation

$$X_a(F) = \int_{-\infty}^{\infty} x_a(t)e^{-j2\pi Ft}\,dt \tag{4.2.72}$$

whereas the signal $x_a(t)$ can be recovered from its spectrum by the inverse Fourier transform

$$x_a(t) = \int_{-\infty}^{\infty} X_a(F)e^{j2\pi Ft}\,dF \tag{4.2.73}$$

Note that utilization of all frequency components in the infinite frequency range $-\infty < F < \infty$ is necessary to recover the signal $x_a(t)$ if the signal $x_a(t)$ is not bandlimited.

The spectrum of a discrete-time signal $x(n)$, obtained by sampling $x_a(t)$, is given by the Fourier transform relation

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \tag{4.2.74}$$

or, equivalently,

$$X(f) = \sum_{n=-\infty}^{\infty} x(n)e^{-j2\pi fn} \tag{4.2.75}$$

The sequence $x(n)$ can be recovered from its spectrum $X(\omega)$ or $X(f)$ by the inverse transform

$$\begin{aligned} x(n) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\omega)e^{j\omega n}\,d\omega \\ &= \int_{-1/2}^{1/2} X(f)e^{j2\pi f n}\,df \end{aligned} \tag{4.2.76}$$

In order to determine the relationship between the spectra of the discrete-time signal and the analog signal, we note that periodic sampling imposes a relationship between the independent variables t and n in the signals $x_a(t)$ and $x(n)$, respectively. That is.

$$t = nT = \frac{n}{F_s} \tag{4.2.77}$$

This relationship in the time domain implies a corresponding relationship between the frequency variabtes F and f in $X_a(F)$ and $X(f)$. respectively.

Indeed. substitution of **(4.2.77)** into **(4.2.73)** yields

$$x(n) \equiv x_a(nT) = \int_{-\infty}^{\infty} X_a(F)e^{j2\pi nF/F_s}\,dF \tag{4.2.78}$$

If we compare (4.2.76) with (4.2.78), we conclude that

$$\int_{-1/2}^{1/2} X(f)e^{j2\pi f n}\,df = \int_{-\infty}^{\infty} X_a(F)e^{j2\pi nF/F_s}\,dF \tag{4.2.79}$$

From the development in Chapter 1 we know that periodic sampling imposes a relationship between the frequency variables F and f of the corresponding analog and discrete-time signals, respectively. That is,

$$f = \frac{F}{F_s} \tag{4.2.80}$$

With the aid of (4.2.80), we can make a simple change in variable in (4.2.79), and obtain the result

$$\frac{1}{F_s}\int_{-F_s/2}^{F_s/2} X\left(\frac{F}{F_s}\right) e^{j2\pi nF/F_s}\,dF = \int_{-\infty}^{\infty} X_a(F)e^{j2\pi nF/F_s}\,dF \tag{4.2.81}$$

We now turn our attention to the integral on the right-hand side of (4.2.81). The integration range of this integral can be divided into an infinite number of intervals of width F,. Thus the integral over the infinite range can be expressed as a sum of integrals, that is,

$$\int_{-\infty}^{\infty} X_a(F)e^{j2\pi nF/F_s}\,dF = \sum_{k=-\infty}^{\infty} \int_{(k-1/2)F_s}^{(k+1/2)F_s} X_a(F)e^{j2\pi nF/F_s}\,dF \tag{4.2.82}$$

We observe that $X_a(F)$ in the frequency interval $(k-\frac{1}{2})F_s$ to $(k+\frac{1}{2})F_s$ is identical to $X_a(F-kF_s)$ in the interval $-F_s/2$ to $F_s/2$. Consequently,

$$\sum_{k=-\infty}^{\infty}\int_{(k-1/2)F_s}^{(k+1/2)F_s} X_a(F)e^{j2\pi nF/F_s}\,dF = \sum_{k=-\infty}^{\infty}\int_{-F_s/2}^{F_s/2} X_a(F-kF_s)e^{j2\pi nF/F_s}\,dF$$
$$= \int_{-F_s/2}^{F_s/2}\left[\sum_{k=-\infty}^{\infty} X_a(F-kF_s)\right]e^{j2\pi nF/F_s}\,dF \tag{4.2.83}$$

where we have used the periodicity of the exponential. namely,

$$e^{j2\pi n(F+kF_s)/F_s} = e^{j2\pi nF/F_s}$$

Comparing (4.2.83), (4.2.82), and (4.2.81), we conclude that

$$X\left(\frac{F}{F_s}\right) = F_s\sum_{k=-\infty}^{\infty} X_a(F-kF_s) \tag{4.2.84}$$

or. equivalently.

$$X(f) = F_s\sum_{k=-\infty}^{\infty} X_a[(f-k)F_s] \tag{4.2.85}$$

This is the desired relationship between the spectrum $X(F/F_s)$ or $X(f)$ of the discrete-time signal and the spectrum $X_a(F)$ of the analog signal. The right-hand side of (4.2.84) or (4.2.85) consists of a periodic repetition of the scaled spectrum $F_sX_a(F)$ with period F_s. This periodicity is necessary because the spectrum $X(f)$ or $X(F/F_s)$ of the discrete-time signal is periodic with period f,, = 1 or F„ = F,.

For example, suppose that the spectrum of a band-limited analog signal is as shown in Fig. 4.18(a). The spectrum is zero for $|F| \geq B$. NOH'. if the sampling frequency F_s is selected to be greater than 2B. the spectrum $X(F/F_s)$ of the discrete-time signal will appear as shown in Fig. 4.18(b). Thus. if the sampling frequency F_s is selected such that $F_s \geq 2B$, where $2B$ is the Nyquist rate. then

$$X\left(\frac{F}{F_s}\right) = F_sX_a(F) \qquad |F| \leq F_s/2 \tag{4.2.86}$$

In this case there is no aliasing and therefore, the spectrum of the discrete-time signal is identical (within the scale factor F_s) to the spectrum of the analog signal. within the fundamental frequency range $|F| \leq F_s/2$ or $|f| \leq \frac{1}{2}$.

On the other hand, if the sampling frequency F_s is selected such that $F_s <$ 28, the periodic continuation of $X_a(F)$ results in spectral overlap, as illustrated in Fig. 4.18(c) and (d). Thus the spectrum $X(F/F_s)$ of the discrete-time signal contains aliased frequency components of the analog signal spectrum $X_a(F)$. The end result is that the aliasing which occurs prevents us from recovering the original signal $x_a(t)$ from the samples.

Given the discrete-time signal $x(n)$ with the spectrum $X(F/F_s)$, as illustrated in Fig. 4.18(b), with no aliasing, it is now possible to reconstruct the original analog

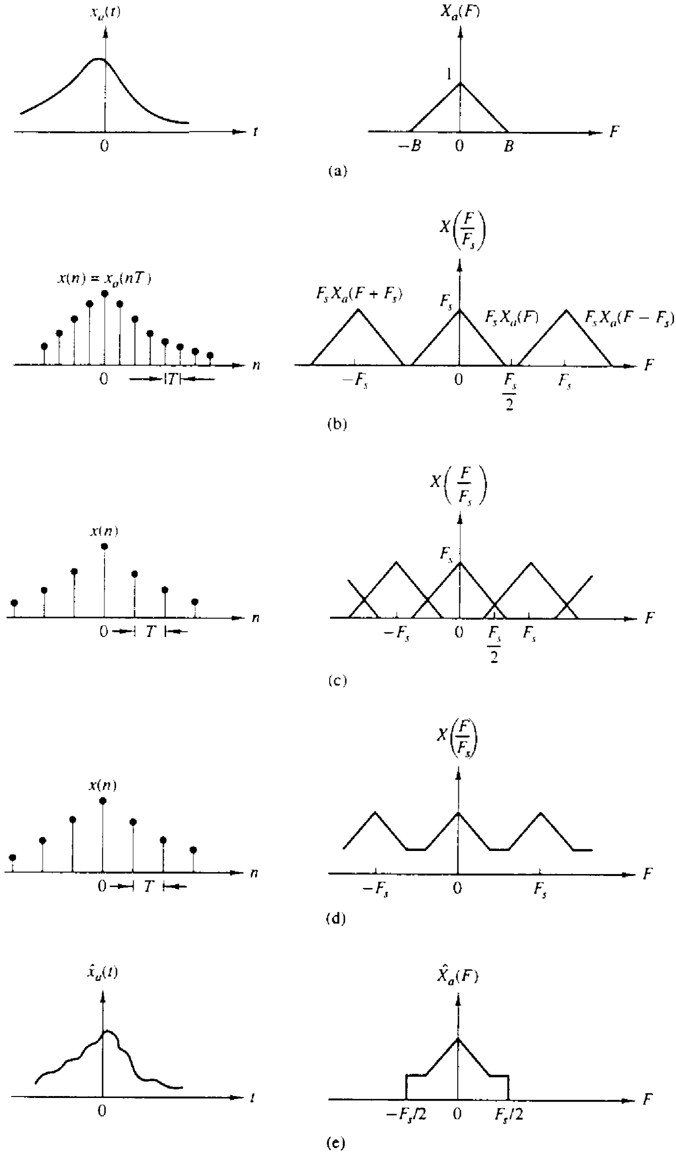


Figure 4.18 Sampling of an analog bandlimited signal and aliasing of spectral components.

signal from the samples $x(n)$. Since in the absence of aliasing

$$X_a(F) = \begin{cases} \dfrac{1}{F_s} X\left(\dfrac{F}{F_s}\right), & |F| \le F_s/2 \\ 0, & |F| > F_s/2 \end{cases} \tag{4.2.87}$$

and by the Fourier transform relationship (4.2.75),

$$X\left(\frac{F}{F_s}\right) = \sum_{n=-\infty}^{\infty} x(n) e^{-j2\pi Fn/F_s} \tag{4.2.88}$$

the inverse Fourier transform of $X_a(F)$ is

$$x_a(t) = \int_{-F_s/2}^{F_s/2} X_a(F) e^{j2\pi Ft}\, dF \tag{4.2.89}$$

Let us assume that $F_s = 2B$. With the substitution of (4.2.87) into (4.2.89), we have

$$\begin{aligned} x_a(t) &= \frac{1}{F_s} \int_{-F_s/2}^{F_s/2} \left[\sum_{n=-\infty}^{\infty} x(n) e^{-j2\pi Fn/F_s} \right] e^{j2\pi Ft}\, dF \\ &= \frac{1}{F_s} \sum_{n=-\infty}^{\infty} x(n) \int_{-F_s/2}^{F_s/2} e^{j2\pi F(t-n/F_s)}\, dF \\ &= \sum_{n=-\infty}^{\infty} x_a(nT) \frac{\sin(\pi/T)(t-nT)}{(\pi/T)(t-nT)} \end{aligned} \tag{4.2.90}$$

where $x(n) = x_a(nT)$ and where $T = 1/F_s = 1/2B$ is the sampling interval. This is the reconstruction formula given by (1.4.24) in our discussion of the sampling theorem.

The reconstruction formula in (4.2.90) involves the function

$$g(t) = \frac{\sin(\pi/T)t}{(\pi/T)t} = \frac{\sin 2\pi Bt}{2\pi Bt} \tag{4.2.91}$$

appropriately shifted by nT, $n = 0, \pm 1, \pm 2, \ldots$, and multiplied or weighted by the corresponding samples $x_a(nT)$ of the signal. We call (4.2.90) an interpolation formula for reconstructing $x_a(t)$ from its samples. and $g(t)$, given in (4.2.91), is the interpolation function. We note that at $t = kT$, the interpolation function $g(t - nT)$ is zero except at $k = n$. Consequently, $x_a(t)$ evaluated at $t = kT$ is simply the sample $x_a(kT)$. At all other times the weighted sum of the time shifted versions of the interpolation function combine to yield exactly $x_a(t)$. This combination is illustrated in Fig. 4.19.

The formula in (4.2.90) for reconstructing the analog signal $x_a(t)$ from its samples is called the ideal *interpolation* formula. It forms the basis for the *sampling theorem,* which can be stated as follows.

Sampling Theorem. A bandlimited continuous-time signal, with highest frequency (bandwidth) B Hertz. can be uniquely recovered from its samples provided that the sampling rate $F_s \ge 2B$ samples per second.

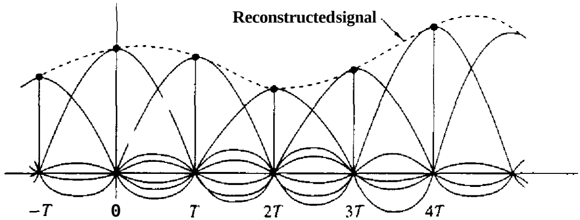


Figure 4.19 **Reconstruction of a continuous-time signal using ideal interpolation**

According to the sampling theorem and the reconstruction formula in (4.2.90), the recovery of $x_a(t)$ from its samples $x(n)$, requires an infinite number of Samples. However, in practice we use a finite number of samples of the signal and deal with finite-duration signals. As a consequence, we are concerned only with reconstructing a finite-duration signal from a finite number of samples.

When aliasing occurs due to too low a sampling rate, the effect can be described by a multiple folding of the frequency axis of the frequency variable F for the analog signal. Figure 4.20(a) shows the spectrum $X_a(F)$ of an analog signal. According to (4.2.84), sampling of the signal with a sampling frequency F_s results in a periodic repetition of $X_a(F)$ with period F_s. If $F_s < 2B$, the shifted replicas of $X_a(F)$ overlap. The overlap that occurs within the fundamental frequency range $-F_s/2 \leq F \leq F_s/2$, is illustrated in Fig. 4.20(b). The corresponding spectrum of the discrete-time signal within the fundamental frequency range, is obtained by adding all the shifted portions within the range $|f| \leq \frac{1}{2}$, to yield the spectrum shown in Fig. 4.20(c).

A careful inspection of Fig. 4.20(a) and (b) reveals that the aliased spectrum in Fig. 4.20(c) can be obtained by folding the original spectrum like an accordian with pleats at every odd multiple of $F_s/2$. Consequently, the frequency $F_s/2$ is called the *folding frequency,* as indicated in Chapter 1. Clearly, then, periodic sampling automatically forces a folding of the frequency axis of an analog signal at odd multiples of $F_s/2$, and this results in the relationship $F = f F_s$ between the frequencies for continuous-time signals and discrete-time signals. Due to the folding of the frequency axis, the relationship $F = f F_s$ is not truly linear. but piecewise linear, to accommodate for the aliasing effect. This relationship is illustrated in Fig. 4.21.

If the analog signal is bandlimited to $B \leq F_s/2$, the relationship between f and F is linear and one-to-one. In other words, there is no aliasing. In practice, prefiltering with an antialiasing filter is usually employed prior to sampling. This ensures that frequency components of the signal above $F \geq B$ are sufficiently attenuated so that, if aliased, they cause negligible distortion on the desired signal.

The relationships among the time-domain and frequency-domain functions $x_a(t)$, $x(n)$, $X_a(F)$, and $X(f)$ are summarized in Fig. 4.22. The relationships for

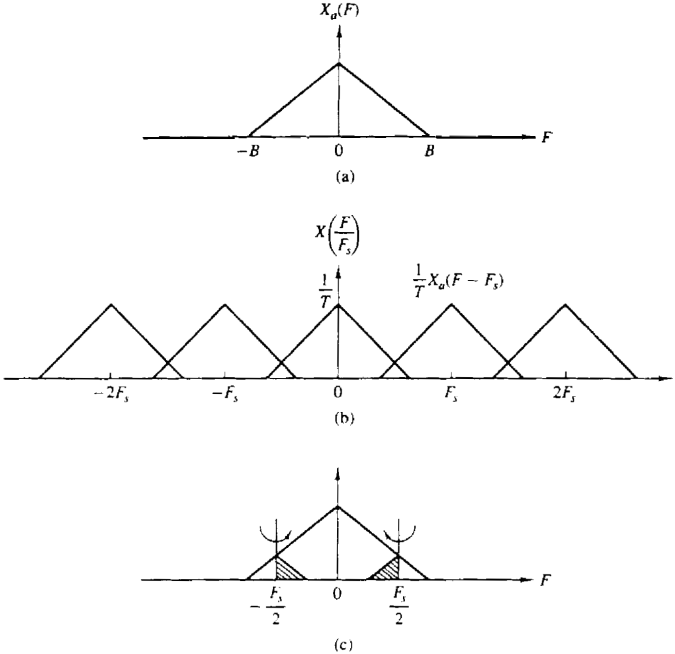


Figure 4.20 Illustration of aliasing around the foiding frequency

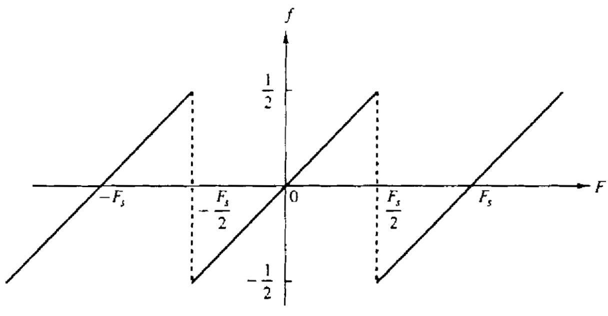


Figure 4.21 Relationship between frequency variables F and f.

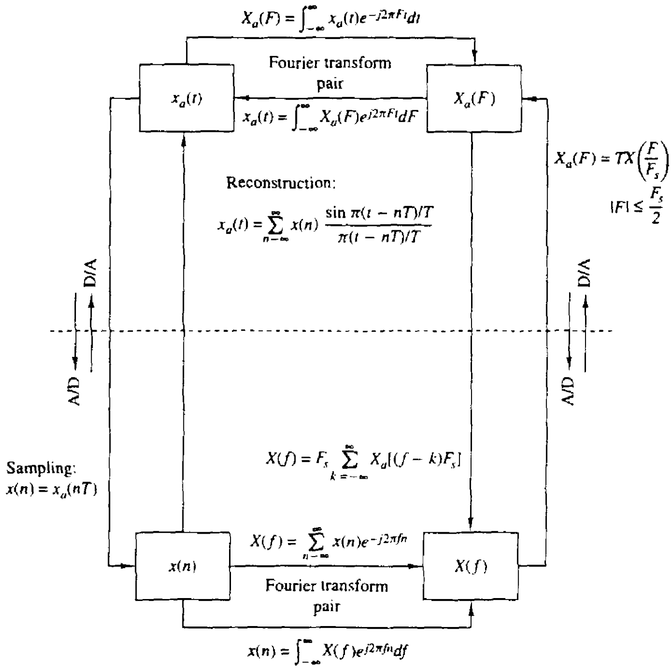


Figure 4.22 **Time-domain and frequency-domain relationships for sampled signals.**

recovering the continuous-time functions, $x_a(t)$ and $X_a(F)$, from the discrete-time quantities $x(n)$ and $X(f)$, assume that the analog signal is bandlimited and that it is sampled at the Nyquist rate (or faster).

The following examples serve to illustrate the problem of the aliasing of frequency components.

Example 4.2.6 Aliasing in Sinusoidal Signals

The continuous-time signal

$$x_a(t) = \cos 2\pi F_0 t = \tfrac{1}{2}e^{j2\pi F_0 t} + \tfrac{1}{2}e^{-j2\pi F_0 t}$$

has a discrete spectrum with spectral lines at $F = \pm F_0$, as shown in Fig. 4.23(a). The process of sampling this signal with a sampling frequency F_s introduces replicas of the spectrum about multiples of F_s. This is illustrated in Fig. 4.23(b) for $F_s/2 < F_0 < F_s$.

To reconstruct the continuous-time signal, we should select the frequency components inside the fundamental frequency range $|F| \leq F_s/2$. The resulting spectrum

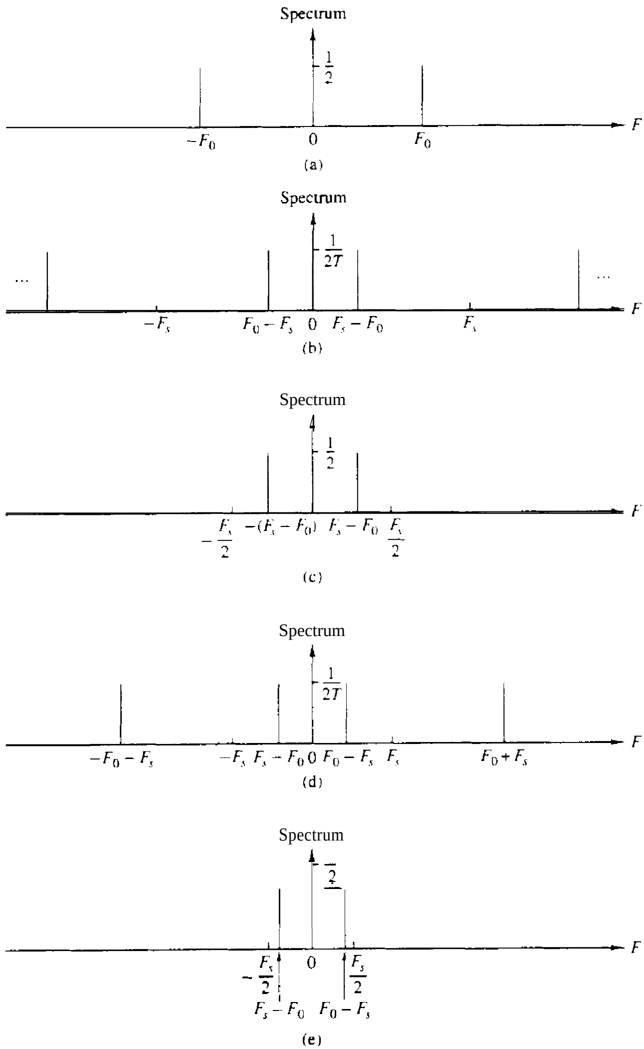


Figure 4.23 **Aliasing** of sinusoidal signals.

is shown in Fig. 4.23(c). The reconstructed signal is

$$x_a(t) = \cos 2\pi (F_s - F_0)t$$

Now. if F_s is selected such that $F_s < F_0 < 3F_s/2$, the spectrum of the sampled signal is shown in Fig. 4.23(d). The reconstructed signal, shown in Fig. 4.23(e), is

$$x_a(t) = \cos 2\pi (F_0 - F_s)t$$

In both cases. aliasing has occurred, so that the frequency of the reconstructed signal is an aliased version of the frequency of the original signal.

Example 4.2.7 Sampling a Nonbandlimited Signal

Consider the continuous-time signal

$$x_a(t) = e^{-A|t|} \qquad A > 0$$

whose spectrum is given by

$$X_a(F) = \frac{2A}{A^2 + (2\pi F)^2}$$

Determine the spectrum of the sampled signal $x(n) \equiv x_a(nT)$.

Solution If we sample $x_a(t)$ with a sampling frequency $F_s = 1/T$, we have

$$x(n) = x_a(nT) = e^{-AT|n|} = (e^{-AT})^{|n|} \qquad -\infty < n < \infty$$

The spectrum of $x(n)$ can he found easily if we use a direct computation of the Fourier transform. We find that

$$X\left(\frac{F}{F_s}\right) = \frac{1 - e^{-2AT}}{1 - 2e^{-AT}\cos 2\pi FT + e^{-2AT}} \qquad T = \frac{1}{F_s}$$

Clearly. since $\cos 2\pi FT = \cos 2\pi (F/F_s)$ is periodic with period F_s, so is $X(F/F_s)$.

Since $X_a(F)$ is not bandlimited. aliasing cannot be avoided. The spectrum of the reconstructed signal $\hat{x}_a(t)$ is

$$\hat{X}_a(F) = \begin{cases} TX\left(\dfrac{F}{F_s}\right), & |F| \le \dfrac{F_s}{2} \\ 0, & |F| > \dfrac{F_s}{2} \end{cases}$$

Figure 4.24(a) shows the original signal $x_a(t)$ and its spectrum $X_a(F)$ for $A = 1$. The sampled signal $x(n)$ and its spectrum $X(F/F_s)$ are shown in Fig. 4.24(b) for $F_s = 1$ **Hz.** The aliasing distortion is clearly noticeable in the frequency domain. The reconstructed signal $\hat{x}_a(t)$ is shown in Fig. 4.24(c). The distortion due to aliasing can be reduced significantly hy increasing the sampling rate. For example, Fig. 4.24(d) illustrates the reconstructed signal corresponding to a sampling rate $F_s = 20$ **Hz.** It is interesting to note that in every case $x_a(nT) = \hat{x}_a(nT)$, but $x_a(t) \neq \hat{x}_a(t)$ at *other* values of time.

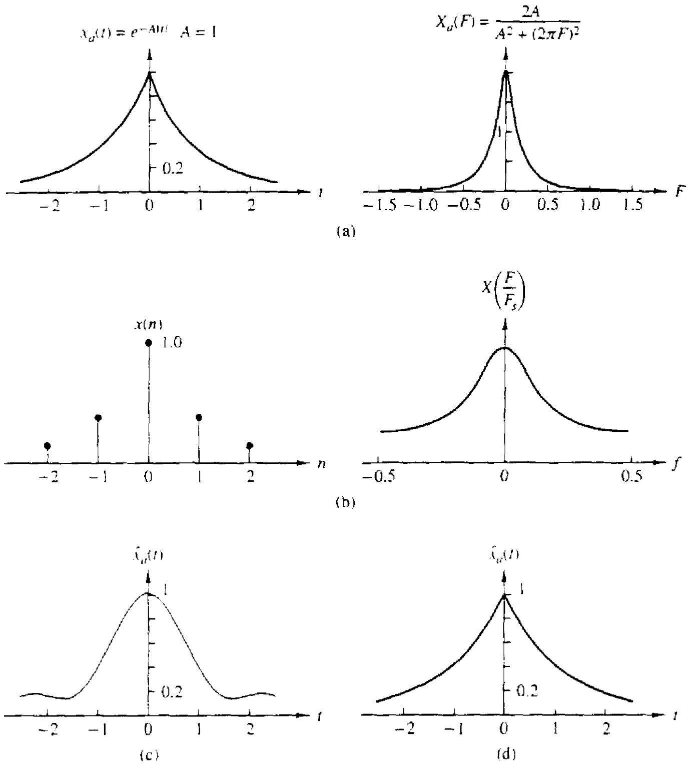


Figure 4.24 (a) Analog signal $x_a(t)$ and its spectrum $X_a(F)$; (b) $x(n) = x_a(nT)$ and the spectrum of $x(n)$ for $A = 1$ and $F_s = 1$ Hz. (c) reconstructed signal $\hat{x}_a(t)$ for $F_s = 1$ Hz; (d) reconstructed signal $\hat{x}_a(t)$ for $F_s = 20$ Hz.

4.2.10 Frequency-Domain Classification of Signals: The Concept of Bandwidth

Just as we have classified signals according to their time-domain characteristics. it is also desirable to classify signals according to their frequency-domain characteristics. It is common practice to classify signals in rather broad terms according to their frequency content.

In particular. if a power signal (or energy signal) has its power density spectrum (or its energy density spectrum) concentrated about zero frequency. such a signal is called a ***low-frequency signal.*** Figure 4.25(a) illustrates the spectral characteristics of such a signal. On the other hand, if the signal power density

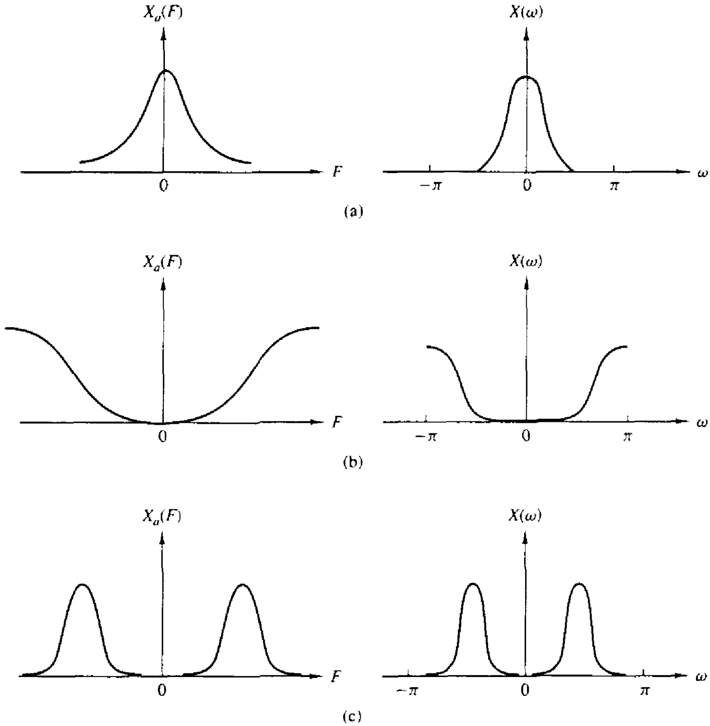


Figure 4.25 (a) Low-frequency. (b) high-frequency. and (c) medium-frequency signals.

spectrum (or the energy density spectrum) is concentrated at high frequencies, the signal is called a ***high-frequency signal.*** Such *a* signal spectrum is illustrated in Fig. 4.25(b). A signal having a power density spectrum (or an energy density spectrum) concentrated somewhere in the broad frequency range between low frequencies and high frequencies is called a ***medium-frequency signal*** or a ***bandpass signal.*** Figure 4.25(c) illustrates such a signal spectrum.

In addition to this relatively broad frequency-domain classification of signals, it is often desirable to express quantitatively the range of frequencies over which the power or energy density spectrum is concentrated. This quantitative measure is called the ***bandwidrh*** of a signal. For example, suppose that a continuous-time signal has 95% of its power (or energy) density spectrum concentrated in the frequency range $F_1 \leq F \leq F_2$. Then the 95% bandwidth of the signal is $F_2 - F_1$. In a similar manner, we may define the 75% or 90% or 99% bandwidth of the signal.

In the case of a bandpass signal. the term ***narrowband*** is used to describe the signal if its bandwidth $F_2 - F_1$ is much smaller (say, by a factor of 10 or more) than the median frequency $(F_2 + F_1)/2$. Otherwise. the signal is called ***wideband.***

We shall say that a signal is ***bandlimired*** if its spectrum is zero outside the frequency range $|F| \geq B$. For example, a continuous-time finite-energy signal $x(t)$ is bandlimited if its Fourier transform $X(F) = 0$ for $|F| > B$. **A** discrete-time finite-energy signal $x(n)$ is said to be ***(periodically) bandlimired*** if

$$|X(\omega)| = 0 \qquad \text{for } \omega_0 < |\omega| < \pi$$

Similarly. a periodic continuous-time signal $x_p(t)$ is periodically bandlimited if its Fourier coefficients $c_k = 0$ for $|k| > M$, where M is some positive integer. A periodic discrete-time signal with fundamental period N is periodically bandlimited if the Fourier coefficients $c_k = 0$ for $k_0 < |k| < N$. Figure 4.26 illustrates the four types of bandlimited signals.

By exploiting the duality between the frequency domain and the time domain, we can provide similar means for characterizing signals in the time domain. In particular. a signal $x(t)$ will be called ***rime-limited*** if

$$x(t) = 0 \qquad |t| > \tau$$

If the signal is periodic with period T_p, it will be called ***periodically rime-limited*** if

$$x_p(t) = 0 \qquad \tau < |t| < T_p/2$$

If we have a discrete-time signal $x(n)$ of finite duration, that is,

$$x(n) = 0 \qquad |n| > N$$

it is also called time-limited. When the signal is periodic with fundamental period N, it is said to be periodically time-limited if

$$x(n) = 0 \qquad n_0 < |n| < N$$

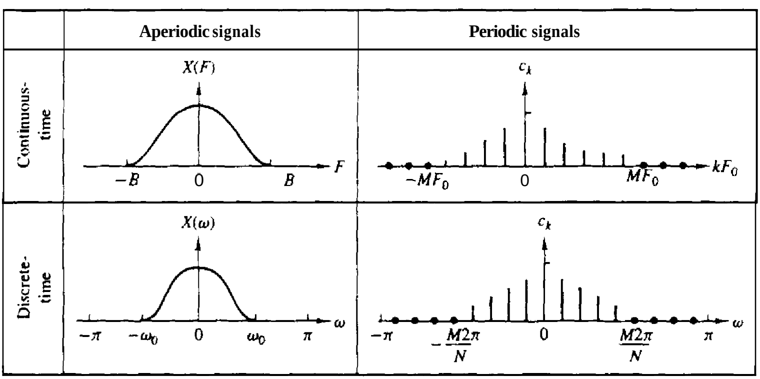


Figure 4.26 Some examples of bandlimited signals.

We state, without proof, that no ***signal*** can be ***time-limited and bandlimited simultaneously.*** Furthermore, a reciprocal relationship exists between the time duration and the frequency duration of a signal. To elaborate, if we have a short-duration rectangular pulse in the time domain, its spectrum has a width that is inversely proportional to the duration of the time- domain pulse. The narrower the pulse becomes in the time domain, the larger the bandwidth of the signal becomes. Consequently, the product of the time duration and the bandwidth of a signal cannot be made arbitrarily small. A short-duration signal has a large bandwidth and a small bandwidth signal has a long duration. Thus, for any signal, the time–bandwidth product is fixed and cannot be made arbitrarily small.

Finally, we note that we have discussed frequency analysis methods for periodic and aperiodic signals with finite energy. However, there is a family of deterministic aperiodic signals with finite power. These signals consist of a linear superposition of complex exponentials with nonharmonically related frequencies, that is,

$$x(n) = \sum_{k=1}^{M} A_k e^{j\omega_k n}$$

where $\omega_1, \omega_2, \ldots, \omega_M$ are nonharmanically related. These signals have discrete spectra but the distances among the lines are nonharmonically related. Signals with discrete nonharmonic spectra are sometimes called quasi-periodic.

4.2.11 The Frequency Ranges of Some Natural Signals

The frequency analysis tools that we have developed in this chapter are usually applied to a variety of signals that are encountered in practice (e.g., seismic, biological, and electromagnetic signals). In general. the frequency analysis is performed for the purpose of extracting information from the observed signal. For example, in the case of biological signals, such as an ECG signal, the analytical tools are used to extract information relevant for diagnostic purposes. In the case of seismic signals, we may be interested in detecting the presence of a nuclear explosion or in determining the characteristics and location of an earthquake. An electromagnetic signal, such as a radar signal reflected from an airplane, contains information on the position of the plane and its radial velocity. These parameters can be estimated from observation of the received radar signal.

In processing any signal for the purpose of measuring parameters or extracting other types of information, one must know approximately the range of frequencies contained by the signal. For reference, Tables 4.1, 4.2, and 4.3 give approximate limits in the frequency domain for biological, seismic, and electromagnetic signals.

4.2.12 Physical and Mathematical Dualities

In the previous sections of the chapter we have introduced several methods for the frequency analysis of signals. Several methods were necessary to accommodate the

TABLE 4.1 FREQUENCY RANGES OF SOME BIOLOGICAL SIGNALS

Type of Signal	Frequency Range (Hz)
Electroretinogram[a]	0–20
Electronystapmogram[b]	0–20
Pneumogram[c]	0–40
Electrocardiogram (ECG)	0–100
Electroencephalogram (EEG)	0–100
Electromyogram[d]	10–200
Sphygmomanogram[e]	0–200
Speech	100–4000

[a] A graphic recording of retina characteristics.
[b] A graphic recording of involuntary movement of the eyes.
[c] A graphic recording of respiratory activity.
[d] A graphic recording of muscular action. such as muscular contraction.
[e] A recording of blood pressure.

TABLE 4.2 FREQUENCY RANGES OF SOME SEISMIC SIGNALS

Type of Signal	Frequency Range (Hz)
Wind noise	100–1000
Seismic exploration signals	10–100
Earthquake and nuclear explosion signals	0.01–10
Seismic noise	0.1–1

TABLE 4.3 FREQUENCY RANGES OF ELECTROMAGNETIC SIGNALS

Type of Signal	Wavelength (m)	Frequency Range (Hz)
Radio broadcast	10^4–10^2	3×10^4–3×10^6
Shortwave radio signals	10^2–10^{-2}	3×10^6–3×10^{10}
Radar. satellite communications. space communications. common-carrier mtcrowavc	1–10^{-2}	3×10^8–3×10^{10}
Infrared	10^{-3}–10^{-6}	3×10^{11}–3×10^{14}
Visible light	3.9×10^{-7}–8.1×10^{-7}	3.7×10^{14}–7.7×10^{14}
Ultraviolet	10^{-7}–10^{-8}	3×10^{15}–3×10^{16}
Gamma rays and x-rays	10^{-9}–10^{-10}	3×10^{17}–3×10^{18}

different types of signals. To summarize, the following frequency analysis tools have been introduced:

1. The Fourier series for continuous-time periodic signals.
2. The Fourier transform for continuous-time aperiodic signals.
3. The Fourier series for discrete-time periodic signals.
4. The Fourier transform for discrete-time aperiodic signals.